



Identifying Progression-based Subtypes of Huntington's Disease: Unsupervised Learning with Clinical Data

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Glossary

Biomarker: Any substance, structure or process that can be measured in the body or its products and can influence or predict the incidence of outcome or disease (World Health Organisation, 2011). Examples would be measurements taken from MRI scans, blood tests or heart rate monitors.

Clinical test score: Non-biological patient-responses to tests administered by clinicians and designed to measure disease. Examples would be a patient's verbal response to questions in a memory test, or physical responses in a motor functioning test.

Disease feature: Refers to *biomarkers* and *clinical test scores* collectively.

Stage: The stage of a disease as it progresses over time. For example, the staging system for cancer is 1-4 and a stage within that system is stage 3.

Subtype: A subtype of a disease based on certain disease features. Subtypes could be based on single cross-sectional measurement of a disease feature, for example blood sugar level determines whether a person has Type I and Type II diabetes. Subtypes could also be based on a distinct temporal pattern of *changes* in disease features (a disease progression-based subtype). Note, in the context of machine learning applied to healthcare, the terms 'subtype' and 'cluster' are often used interchangeably.

Abstract

Objective: The main objective of this study was to determine whether data-driven subtypes of Huntington's Disease (HD) exist based on motor, cognitive and psychiatric clinical symptoms. A secondary objective was to test whether symptomatic variance existed independently of the biomarker, CAG repeat length.

Methods: A data subset of 5947 patients and controls across 9 features (3 motor, 3 cognitive and 3 psychiatric test scores) from the Enroll-HD data set were used. Analysis of the data included subtyping and stage inference (using the SuStaIn model), MCMC uncertainty estimation, 10 fold cross validation and t-tests between subtypes.

Results: Three subtypes with distinct progression patterns were found: *Motor-led HD* (Subtype 1), *Cognitive-led HD* (Subtype 2) and *Psychiatric-led HD* (Subtype 3). The proportions of subtypes in each group were 0.37, 0.33 and 0.3 respectively. Statistically significant differences were found between Subtype 1 and Subtype 2 and between Subtype 2 and Subtype 3, both with p-values of < 0.01 . Statistically significant differences were not found between the CAG distributions of the subtypes (p-values > 0.3).

Discussion: This is the first time progression-based subtypes of HD have been found using clinical test data. The subtypes discovered could have large implications on HD treatment scheduling, drug discovery design and our general understanding of the disease (for example the *Motor-led HD* subtype contradicts a key existing clinical definition of HD which states that motor symptoms usually come later). Additionally, it was shown that the distinct progression patterns (subtypes) existed independently of CAG repeat length, fortifying the argument that CAG repeat length is not a suitable biomarker for predicting disease progression.

1 Literature review

1.1 Introduction to Huntington's Disease (HD)

Huntington's disease (HD) is a hereditary, progressive neurodegenerative disease for which there is currently no cure. HD occurs in 5-10 in 100,000 people in Western Europe, several Americas countries and Oceania, and drops dramatically to 0.1-0.5 in 100,000 in most Asian and African countries (Cepeda & Xiao-Ping, 2018).

HD is characterised by a triad of symptoms: progressive motor dysfunction (i.e. involuntary twitching and writhing), cognitive decline (i.e. memory retrieval, executive functioning), and psychiatric disturbance (i.e. depression, anxiety and irritability), (Tabrizi et al, 2009).

According to the Diagnostic & Statistical Manual of Mental Disorders (DSM IV), the onset of motor abnormalities tends to occur in the third or fourth decades of life, however cognitive and psychiatric symptoms can occur much earlier than this. HD is equally prevalent in men and women, and people with HD have a 50% chance of passing it onto their offspring (DSM IV).

Clinical test scores, which measure symptoms, are important in the study of HD, partly because they are used in diagnosing the onset of manifest HD, but also because reducing symptoms is a main clinical goal in drugs trials. In other words, the end-goal of most HD drugs is to stop or slow down motor, cognitive and psychiatric decline.

To understand the drivers behind the symptoms, it's important to understand the genetics and neuropathology of the disease. These will be discussed next.

1.2 Genetics of HD

The most important biomarker of HD is the CAG repeat length which is measured using a blood sample and used to determine whether a person has HD. The CAG trinucleotide repeat is a DNA segment within the HTT gene. It is made of three building blocks that appear multiple times in a row on the short arm of the chromosome 4 (Hoogeveen et al., 1993). Normally this CAG (cytosine-adenine-guanine) segment is repeated 10-35 times within the

chromosome. In HD patients it is repeated 36 to 120 times. People with 36-39 repeats may not exhibit symptoms, but those with repeats above 40 almost always do (US National Library of Medicine, 2020). The HTT gene, which contains the CAG repeat, plays the role of providing instructions to make huntingtin protein (US National Library of Medicine, 2020). An expanded CAG repeat causes an expanded polyglutamine chain in the huntingtin protein (Hoogeveen et al., 1993), resulting in an abnormally long protein. This elongated protein manifests as the mutation of the ubiquitously expressed HTT gene which causes HD (Möller, 2012). The elongated huntingtin protein is then cut into smaller, toxic fragments that bind together and accumulate in neurons, severely disrupting the normal functions of these cells. This process particularly affects regions of the brain that help coordinate movement, and those which control thinking and emotions (US National Library of Medicine, 2020).

A highly abstracted and simplified version of the chain of effect from gene mutation to symptomatology can be seen in Figure 1 below.

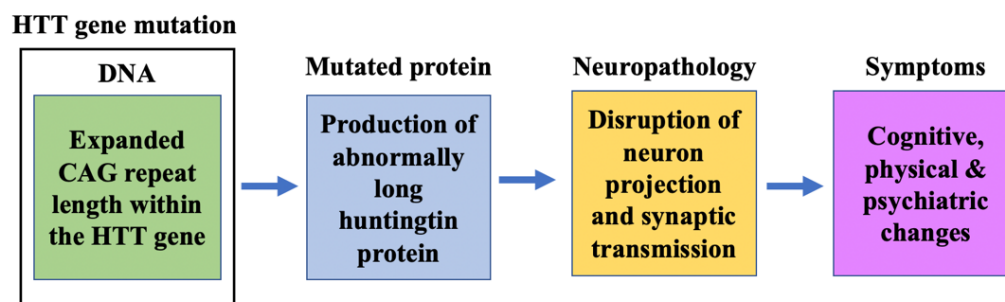


Figure 1: The chain of biological effect from CAG repeat length to symptoms. (Image formed from information in Bates et al, 2014; US National Library of Medicine, 2020; Hoogeveen et al.,1993; Möller, 2012)

1.3 Neuropathology of HD

Neurodegeneration is the root cause of symptoms in HD (Bates et al, 2014) therefore it is useful to understand the neurobiology that underpins the disease. This section will explain the link between neuropathology and the motor, cognitive and psychiatric symptoms associated with HD.

The main brain regions affected by HD are the neo-striatum in the basal ganglia, and the cerebral cortex. However, other regions of the basal ganglia can also be affected, especially

those which receive projections from the striatum; namely, the globus pallidus and substantia nigra, (Bates et al, 2014).

1.3.1 Striatum

The striatum (made up of the caudate nucleus and putamen) is a key component in the motor and reward systems, and as such, pathology in this region is the most likely cause of motor symptoms in HD. The striatum works by receiving glutamatergic and dopaminergic inputs from different sources, and is the primary input to the rest of the basal ganglia (Bates et al, 2014).

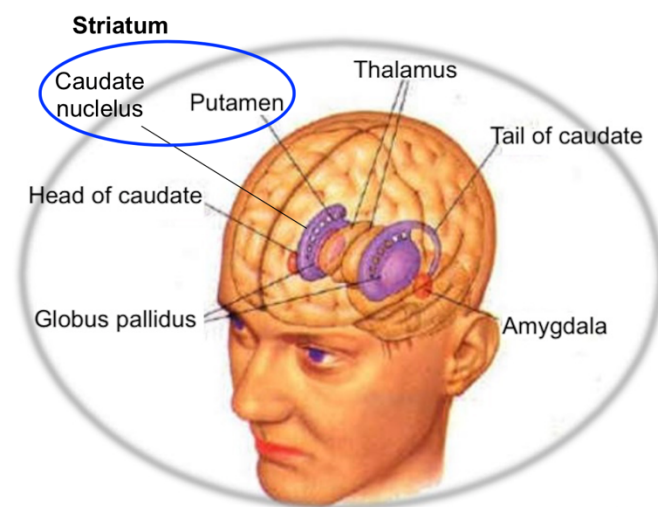


Figure 2: The Basal Ganglia (Dwivedi, 2012)

The striatum comprises two types of neurons; projection neurons and local circuit neurons. The neuropathology of HD is most clearly characterised by loss of projection of neurons from the striatum (Precious et al, 2018). Pathology can also be seen via post-mortem examination by a significant bilateral atrophy in the striatum (de la Monte et al, 1988; Ayleward et al, 1997; Vonsattel and DiFiglia, 1998; Vonsattel et al, 2008).

The volume of the caudate nucleus and putamen (in the striatum) is also reduced in HD patients, with one study showing, respectively, an average of 64% and 57% cross sectional area loss (de la Monte et al, 1988).

1.3.2 Cerebral cortex and other regions of the Basal Ganglia

Cortical atrophy has also been observed in HD patients in the form of volume loss, cortical thinning, changes in neuron and glial size, and neuronal loss (MacDonald & Halliday, 2002; Rosas et al, 2002, 2008; Ruocco, 2008; Thu et al, 2010). Studies have shown averages of 42% neuron loss in the motor cortex and 49% neuron loss in the premotor region in HD patients (MacDonald & Halliday, 2002). MRI studies have also shown thinning in various regions of the cortex, with the primary motor and visual, and sensory cortical regions being most affected (Rosas et al, 2008). This cortical atrophy is likely another key cause of motor-related symptoms associated with HD.

The pars reticula part of the substantia nigra (in the basal ganglia) has also shown loss of neurons in HD patients (Campbell, 1961; Richardson 1990). This area plays a major role in dopamine projection, and it is suggested that loss of these projections could lead to symptoms of HD (Yohrling et al, 2003), likely those in the psychiatric realm.

Other brain regions have also been shown to be affected by HD, albeit to a lesser degree, including the thalamus, hypothalamus, subthalamic nucleus, hippocampus and cerebellum (Reiner et al, 2014).

In summary, HD damages many areas of the brain on both a cortical and subcortical level, affecting brain volume and neurosynaptic transmission in several critical functional areas. It is this damage which causes the myriad of disabling symptoms across cognition, motor functioning and mental health.

1.4 Treatment for HD

There is currently no treatment for modifying the progression of HD, and only very few symptomatic therapeutic options are available (Precious et al, 2018). Clinically approved drugs for treating the symptoms of HD include, for example, those to control movement (such as tetrabenazine and deutetabenazine) and drugs to control psychological symptoms (such as anti-depressants, anti-psychotics and mood stabilizers).

In addition to being symptom-only drugs, all come with the risk of substantial side effects such as obsessive compulsive disorder, depression, psychiatric conditions and even other movement disorders.

There has however been recent research which shows positive signs towards finding a progression-altering therapeutic which is discussed in the next section.

1.4.1 Recent clinical trials & methods

None of the drug trials aiming to alter the progression of HD have so far been successful, except the Tabrizi et al (2019) research in which a drug named RG6042, designed to target the huntingtin gene expression, was developed. The drug targets molecules which play the role of "messenger" between DNA and protein, known as messenger RNA or mRNA (messenger RNA). In the case of HD, mRNA is the conduit between the HTT gene and mutant huntingtin protein. In Phase 1-2a clinical trials, dose-dependent reductions in concentrations of mHTT were observed in patients who were administered RG6042, as was the absence of serious adverse effects. As of January 2020, RG6042 had reached phase 3 clinical trial (the final stage before FDA approval in the USA) which is expected to finish in 2022. This is the furthest stage a drug that targets the root of HD, rather than the symptoms, has reached. It is yet to be determined empirically whether these reductions in mHTT will manifest as reduced symptoms of HD, which is the ultimate goal. However, based on the chain of biological effect (illustrated in Figure 1), the prospect of reduced symptoms as a result of reduced mHTT is highly plausible.

Several other early stage research studies have also opened up entirely new pathways for drug development for HD. For example, Rebec (2018) demonstrated that increased expression of glutamate transport proteins can restore glutamate deficits in the striatum, while normalising striatal neuronal activity and improving behavioural signs. These findings are far from clinical trial stage at the moment, but open up new routes of investigation for future drug targeting.

It is important to discuss the methods used in clinical trials for drugs aimed at slowing down or stopping HD progression. These trials typically take the form of randomized, double-blind,

placebo-controlled trials, where participants may be stratified by stage. For example the Tabrizi et al (2019) study accepted only early stage (clinical stage 1) participants. However, beyond that, little cohort stratification is done. It would seem no trials have been done to date where patients are stratified based on disease progression subtype, or indeed other phenotype-based subtype.

If HD subtypes do exist, ignoring them in clinical trials could have a big impact on the validity results. For example, consider that a subset of HD patients tend to experience the majority of their motor decline early on, and that this period of decline has passed by the start of clinical trial. After the trial, absence of further motor decline is likely due to the nature of the disease subtype. However, the absence of motor decline could be incorrectly attributed to the drug instead. If we knew going into a trial that one subset of HD patients are likely to have passed through the key motor decline phase already, and are now in the cognitive decline phase for example, we would be in a better position to measure the true effect of the drug.

In short, there could be substantial clinical value in measuring the effect of a drug on different HD subtypes, comparing them, and seeing if some subtypes respond to the drug better than others. The existence of such subtypes is evidenced by symptomatic and neurological heterogeneity in the disease, which is discussed in the next section.

1.5 Symptomatic & neurological heterogeneity of HD

One potential reason for so few successes in developing a drug to slow or stop HD progression is the neurological and symptomatic heterogeneity seen in HD patients. Despite the single gene aetiology of HD, highly variable degeneration has been seen in the striatum and cerebral cortex; variation which seems to reflect the high variation in symptoms of HD (Hadzi et al, 2012; Pillai et al, 2012). This gives rise to the hypothesis that gearing clinical trials towards subtypes of HD, rather than HD as a whole, could potentially pave the way for progression-altering drug therapies (as discussed in the previous section).

There are several types of data with which we could try to identify subtypes. The first is CAG repeat length. However, the variation mentioned above has been shown even in cases with the same or similar number of CAG repeats (Georgiou et al, 1999; Friedman et al, 2005; Gome-Esteban et al, 2007; Thu et al, 2010). For example, identical twins, with the same CAG repeat length, have shown significant differences in symptoms (Georgiou et al, 1999; Friedman et al, 2005; Gomez-Esteban et al, 2007). This suggests that whilst CAG repeat length may be useful in predicting *whether* someone will show symptoms it is not necessarily a suitable biomarker predicting *types* of symptoms.

Given neuropathological variance has been shown to be in line with symptomatic variance, independent of CAG repeat length (Hadzi et al, 2012; Pillai et al, 2012), it would therefore make sense to seek out subtypes based on either neurological or symptomatic (clinical) variance in the data.

In this study, clinical test data, which measures the symptoms of motor, cognitive and psychiatric decline, will be used for two reasons. Firstly, the overarching goal of this study is to identify subtypes which could be used for patient stratification in clinical trials, in the hope that this will increase the chances of finding a progression-altering drug. Given that the success of such drug trials tend to be measured by reduction in clinical symptoms, it would make sense to find subtypes based on clinical symptoms. Secondly, large volumes of clinical data have been collected and are readily available for use (more so than neural data).

1.6 Subtyping & stage classification of HD

There have been several efforts to group HD patients into smaller fractions. The next section will summarise the currently-available staging and subtyping systems for HD.

1.6.1 Stage classification of HD

Staging systems can be very useful in disease management. For example, determining where a cancer patient lies between stages 1 and 4 may help doctors decide whether radiotherapy or chemotherapy is needed.

Several traditional disease progression systems exist and are clinically used for the staging of HD. These are based either on neuropathology or symptoms. The Vonsattel Grading System (Vonsattel et al., 1985; Vonsattel & DiFiglia, 1998) for example, is a neuropathological grading system based on the distinctive temporospatial pattern of degeneration in the striatum. Patients are graded by disease stage based on gross and microscopic findings using conventional methods of examination obtained from three standardized, coronal sections (taken by MRI for example) that include the striatum (Bates, 2014). Grade 1 cases for example show neuropathologic changes that can be detected microscopically but without gross atrophy. By grade 4, striatal atrophy is most severe, and the medial surface of the caudate nucleus is concave. A more detailed explanation of this taxonomy can be seen in Appendix 1. A major drawback of this system is that it is based on brain scans which require large and sophisticated technology and as such, can be more expensive, and so more difficult, to collect that clinical data.

A well established capability-based stage grading system is that by Shoulson & Fahn (1979) which measures, general functioning from stages 1 to 5 based on Total Functional Capacity; a functional score in the UHDRS (Unified Huntington's Disease Rating Scale; Huntington's Study Group, 1996). Stage 2 for example can be defined as *'the patient is still functional at work, but at a lower capacity than stage 1'* and last for between 3 and 13 years after onset (UC San Diego School of Medicine, 2020). Stage 3 can be defined as *'the patient can do very little work and requires assistance in household responsibilities'* and can last between 5 and 16 years after onset. (A more detailed version of this staging system can be seen in Appendix 2). The main drawback of this approach is that it only covers general capability. This means the stage classifications hold no information about whether incapability is a result of motor, cognitive or psychiatric decline. Therefore, whilst this classification is indicative of support needed by the patient, it provides very little guidance to clinicians in regards to treatments.

1.6.2 Subtypes of HD

Disease subtyping involves using cross-sectional biodata (such as blood sugar levels, MRI scans or genetic data) or clinical test scores (such as memory tests) to define disease subtypes. These subtypes are typically unrelated to progression.

An example of this kind of subtyping is that by Woodruff et al (2009), who identified a subtype of asthma based on a specific type of disease-fighting white blood cell (called eosinophils). This was discovered using unsupervised hierarchical clustering. It was later found that the subtype uniquely responded to a novel therapy which targeted the specific eosinophil (Corren et al, 2011). This kind of precision medicine can transform the way a disease is treated and demonstrates the value of subtyping.

In relation to HD, whilst significant neurological and symptomatic heterogeneity has been found observationally, which supports the hypothesis that subtypes exist, no formal data-driven subtyping systems have been defined.

As such, seeking out any kind subtypes of HD would be a worthy research pursuit. However, subtyping system that also provided an improved *progression* model, would be especially valuable to the medical community. This prospect is discussed in the next section.

1.7 The gap in knowledge

There exists a need for two separate developments in HD research. The first is a more fine-grained staging system which improves on the traditional systems that already exist. The second is a first-ever Huntington's disease subtyping system.

The need for the latter was identified in the previous section (there is currently no subtyping system for Huntington's disease). The need for an improved staging system is multi-fold. As can be seen in section 1.6.1, all key HD staging systems, whether neurological or functional test-based, have 4-5 stages with some individual stages expected to last for anywhere between 1 and 11 years. Whilst these grading systems are useful in the sense that they give clinicians and patients some idea of what symptoms they should be planning for next, they are somewhat crude. Firstly they are crude in the sense that, despite evidenced heterogeneity, one staging system is used for Huntington's disease as a whole, and does not account for different types of the disease. This means people who suffer from more severe cognitive decline for example, are grouped similarly to those with more aggressive motor decline. Secondly, none give us information specific to motor, cognitive and psychiatric decline;

instead they are based on either neural decay or a decline in general functioning (ability to work for example). Thirdly, the stages in both the neurological and symptomatic taxonomies discussed above were derived semi-arbitrarily based on subjective thresholds defined by the researchers. In these methods, the researchers made an educated yet subjective judgement about what comprises a stage. These stage classifications methods are therefore subject to researcher bias and, by default, can only be based on features which can be seen by the researcher, meaning valuable information could be missed. A means to overcome this would be to derive stages more quantitatively, with the assistance of latent or 'hidden' variables. The data collection and interpretation itself, whether in the form of biomarkers or clinical test scores, would still be subject to human bias and error, but the stages themselves would be largely quantitative. This data-driven approach is discussed further in the next section.

1.7.1 Data-driven disease modelling

In data-driven disease modelling, progression stages are derived from the data, rather than defined by the researcher or clinician. The advantages of this approach are multi-fold. Firstly, because the stages are based on data, they are less exposed to researcher bias. Secondly, hidden (latent) variables in the data can be used to create a more detailed progression model than those based only on observable features. And lastly, if the model takes account of different symptom types (motor, cognitive and psychiatric), it can be used to more accurately plan treatment. More effective clinical scheduling can take place if a clinician knows that a patient is likely to require a certain treatment within a certain time frame. Likewise, anticipation of progression can be used to drive a patient's own behaviour towards recovery. For example, a progression model of Type II diabetes could allow a patient to see how their health could deteriorate over time if they don't change their consumption and activity habits.

In HD, progression models can be especially valuable since patients who carry the mutated HTT gene often have an extended pre-manifest period, during which therapeutic intervention is likely to be most effective (Wijeratne et al, 2018).

Recent studies have been carried out with aim of classifying Huntington's disease progression in this way. A study by Wijeratne et al (2018) on the TRACK-HD dataset for example, found that Huntington's patients could be stratified into disease stage based on the order in which biomarkers transition from normal to abnormal. The data used were structural MRI images

and the biomarkers were brain regional volume differences. The probabilistic Event Based Model (Fonteijn et al, 2012) was used (explained later in sections 1.8.1 and 2.3.1) and the most significant changes were seen in the putamen and caudate.

However, no studies have been seen which aim to model stage progression based on clinical test score (rather than biomarker) data such as functional, psychiatric and cognitive test scores. Additionally, no studies have been done which aim to find subtypes of HD.

This project will aim to do two things. The first is to find an improved staging system for HD based on clinical test scores (from cognitive, motor and psychiatric tests). The second is to, for the first time, find subtypes for HD. It will do so by aiming to find disease subtypes *based on* progression patterns.

To understand the options for doing this, it is important to first understand the current staging and subtyping techniques available. The next two sections will briefly discuss this.

1.8 Data-driven disease modelling: Methods

Many methods of disease modelling exist. Examples are shown in Table 1 below.

Type	Description	Data type	Publication
Machine learning-based	Bayesian Gaussian process classification.	Cross-sectional, Gaussian	Young et al. (2013)
Regression-based	Multivariate nonlinear mixed effects model (for estimating the trajectories of voxel wise neuroimaging biomarkers).	Longitudinal	Bilgel et al. (2016)
Continuous trajectory	Trajectory of features over time.	Longitudinal	Villemagne et al. (2013); Donohue et al. (2014)
Spatiotemporal	Gaussian process-based method (for the joint modelling of imaging and clinical biomarker progressions from time series of individual observations). Non-parametric.	Longitudinal	Lorenzi et al. (2017)
Event based	EBM (Event-based model). Progression is based on disease features transitioning from normal to abnormal.	Cross-sectional, Gaussian	Fonteijn et al. (2012)
Event based	KDEMM (Kernel Density Estimation Mixture Model). Progression is based	Cross-sectional	Frith et al (2018)

	on disease features transitioning from normal to abnormal but for non-gaussian data. Non-parametric.		
Event based	Linear Z-score Model. Progression is based on disease features transitioning between z-scores (deviation from the control mean).	Cross-sectional, Gaussian	Young et al (2018)

Table 1: Disease progression modelling methods

As can be seen in Table 1, some methods can model progression based only on cross-sectional (rather than longitudinal) data. This is particularly important given that in medical research, people are often observed only once. Additionally, and partly in consequence to this, the volume of single-measurement data available is substantially higher than that of longitudinal (whereby several separate measurements are needed and so volumes of data can be affected by factors such as missed appointments, participant drop-out or early termination of the study). Therefore, using a model that can derive a stage taxonomy from cross-sectional data vastly opens up the volume and variety of data which can be used, and thus increases the chances of discovering a valid staging system.

As such, this section will only examine data-driven progression modelling techniques that can work on cross-sectional data.

1.8.1 The EBM (Event Based Model)

The EBM (Fonteiin et al, 2012) describes disease progression as a series of events, S , which it infers from the data, X (the mathematical notation for this can be seen in section 2.3.1). Here, an *event* constitutes a biomarker transitioning from a normal to an abnormal level. To provide a general medical example of this transition, when the biomarker *fasting blood sugar level* reaches 126 mg/dL it is considered abnormal and indicates diabetes (American Diabetes Association, 2020). However, the blood sugar level threshold in this example was decided semi-arbitrarily by clinical researchers. With EBM, the model learns the characteristics of 'abnormal' from the data itself, which are labelled 1 for disease and 0 for control. Having been identified directly from the data, the stages are considered hidden variables.

Parametric mixture modelling is used to calculate the likelihoods of events from the biomarker measurements. Each event then has its own likelihood distribution, as is abstractly illustrated in Figure 3.

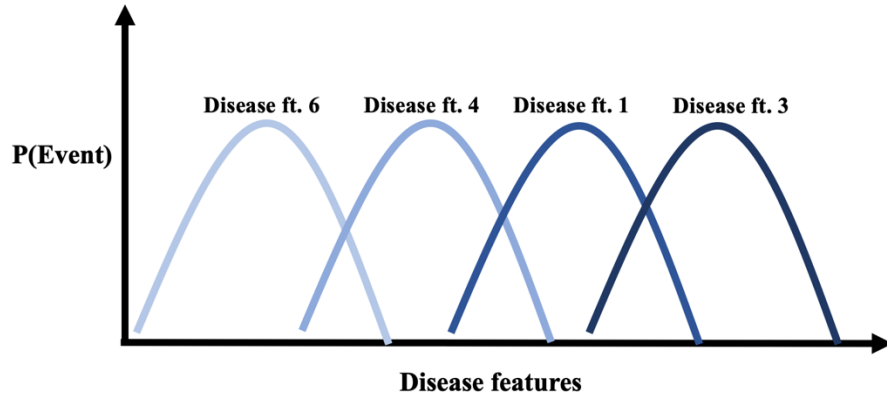


Figure 3: Mixture Model of Event Likelihoods in the EBM

Each disease feature has its own distribution showing the likelihood of it having transitions to abnormal, $P(\text{Event})$

The progression pattern is then defined as the maximum likelihood sequence of events based on the event likelihood distributions (see Figure 4 below).

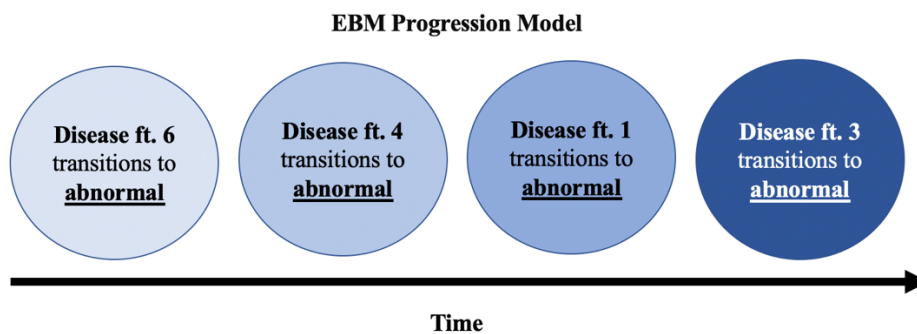


Figure 4: Disease progression defined by the Event Based Model

This figure shows how disease X progresses over time, based on the order in which features of the disease transition from normal to abnormal. (ft. = feature).

There are three key assumptions of the EBM. The first is that measurements decrease monotonically as the disease progresses. The second is that the event sequence is consistent over all patients. The third is that the data is Gaussian.

Recently, the EBM has been used to estimate the order of brain region volume loss in sporadic Alzheimers (Young et al. 2014) and brain region volume loss in Huntington's disease (Wijeratne et al, 2018).

A key advantage of the EBM, in addition to the fact it can work on cross-sectional data, is that it does not require a priori division of the patients into clinical stages. This means the model does not first need to learn what a stage looks like from labelled data (unlike in supervised learning where labelled data is needed in order to make future predictions). Being able to use unlabelled data vastly opens up the volume of data and variety of datasets available. Another advantage is the interpretability of the model which allows us to see clearly how it arrives at its output. This kind of interpretability is critical in medicine where clinical decisions based on the model output could mean life or death and so need to be explainable.

A downside is that, due its binary nature, it provides a somewhat crude model of progression, telling us only whether a biomarker is normal or abnormal, and nothing in between. Secondly, data has to be Gaussian to be used in the EBM, which could limit the type of data it can be used with. Cognitive test data for example, often isn't Gaussian (Firth et al, 2018) and so would yield poor results with the EBM.

1.8.2 The Linear Z-score Model

The Linear Z-score model is an extension of the EBM model with two differences. The first is that the input data is z-scores of the biomarker measurements, rather than raw measurements. The second is that an *event* is now defined as a biomarker having transitioned from one z-score to the next, rather than from normal to abnormal. This is illustrated in Figure 5.

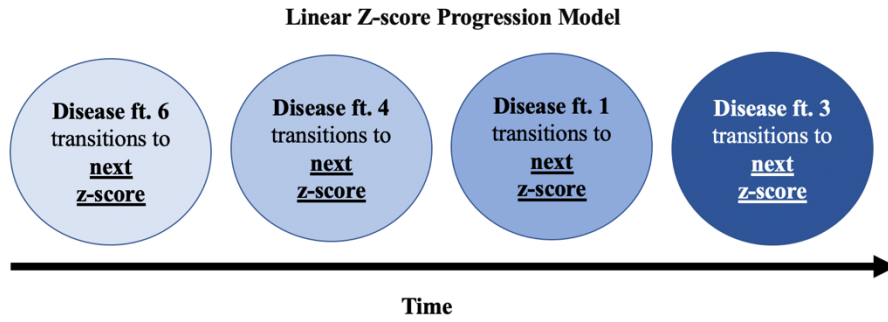


Figure 5: Disease progression defined by the Linear Z-score model

This figure shows how disease Y progresses over time, based on the order in which disease features transition between stages. For example Disease feature 4 could be *fructosamine levels* which at stage 1 = $x \mu\text{mol/L}$ and at stage 2 = $x+3 \mu\text{mol/L}$. (ft. = feature).

The key advantage of the linear z-score model is that it provides a much more fine-grained model of disease progression compared to EBM, since every disease feature can have several events as they transition between different z-scores (rather than just one event like in EBM; the transition to abnormal). A second advantage is that like EBM, its output is explainable.

A downside of Z-Score SuStaIn is that it can only handle Gaussian data (also like EBM). If the variables are non-Gaussian, results will be poor and so a non-parametric method is needed. Secondly, whilst the model is able to derive the maximum likelihood ordering of events, it does not have a measure of time, therefore it does not tell us how long each stage is likely to last.

1.8.3 The Kernel Density Estimation Mixed Model (KDEMM)

KDEMM (Firth et al, 2014) is another extension of the EBM but which was designed specifically to work with non-Gaussian data. The model integrates the non-parametric smoothing technique, Kernel Density Estimation (KDE) to estimate the pdfs of the non-Gaussian random variables. The smoothed pdfs can then be used in the EBM (Mixture Model) to model disease progression.

Due to the nature of the data being non-Gaussian, z-scores cannot be used and so the KDEMM models disease progression the same way as the EBM; based on disease features transitioning from normal to abnormal.

In all three examples discussed here (the EBM, the Z-score model and the KDEMM), subjects are assumed to be samples of a single common progression pattern. That is, we assume all patients' symptoms will progress in the same way. For Huntington's disease where significant symptomatic and neurological heterogeneity is seen, it is unlikely that this is a true representation of the disease. Therefore, if progression patterns themselves can be subtyped, better predictions about what's in store for patients as time passes may be possible. Moreover, subtyping HD by progression could highlight the need for different treatments and pave the way for more precision drug therapies.

Many subtyping (frequently known as "clustering") techniques exist. The next section briefly discusses some key methods available, and then goes into detail on a unique method that is able to subtype based on progression pattern.

1.9 Data-driven subtyping models

1.9.1 The clustering methods landscape

Subtyping, often referred to in machine learning as *clustering*, is an unsupervised learning technique which involves taking a set of unlabelled data, inferring subtypes (clusters) from that data, and allocating each observation to one of those groups. Observations are grouped (clustered) by similarity whereby each observation has more in common with the other observations in their cluster, than they do with observations in other clusters.

Disease clustering, typically done using cross-sectional data, is useful in medicine as it allows clinicians to formulate diagnoses with more specificity, leading to more targeted treatments and interventions (known as precision medicine). A well known general example of disease subtyping is that of diabetes which is split into Type I and Type II (as mentioned in section 1.7.1), where each subtype requires different treatments.

Many clustering techniques are available, the variation between which is typically based on how they define and measure 'similarity'. These techniques tend to fall into three categories; hierarchical, partitional and density based.

Hierarchical algorithms (such as agglomeration) start by considering each data point to be its own cluster (*singletons*). Each cluster is then paired with the most proximal clusters until there is one cluster left. This method works well in finding embedded structures in the data. A downside is that it assumes every observation is relevant when usually it is not, meaning the result can be skewed by noise.

Partitional algorithms (such as k-means and Gaussian Mixture Models) on the other hand start by defining the number of clusters, k , and the number of iterations. This is set before running the algorithm. The algorithm then randomly allocates k *centroids*, and observations closest to the centroid are grouped with it. Partitional algorithms also take account of every data point in the set and as such, can be subject to noise. Additionally, because partitional clustering methods rely on researcher decisions about number of clusters and iterations before running the algorithm, they tend to require more prior knowledge.

Density based algorithms (such as DBSCAN; Density Based Spatial Clustering of Applications with Noise) take the approach of only looking at points which are tightly packed, and assuming the rest is noise. They are good for finding unknown clusters of similar density. However, conversely to hierarchical methods, they risk discarding data which are actually useful, when classifying it as noise.

1.9.2 The SuStaIn (Subtype & Stage Inferencing) model

All of the subtyping methods in the previous section are capable of splitting a disease into cross-sectional subtypes, as is abstractly illustrated in Figure 6 below.

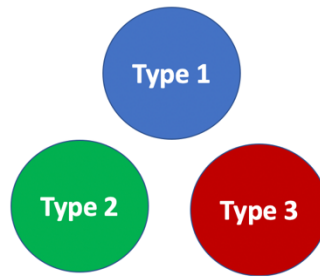


Figure 6: Cross Sectional Subtypes

An example would be splitting Alzheimer's into Type 1, Type 2 and Type 3 based on neurological biomarkers

This is in contrast to the progression models mentioned previously in section 1.8, whereby a single disease type is split into different stages based on the evolution of disease features over time. This is illustrated in Figure 7 below.

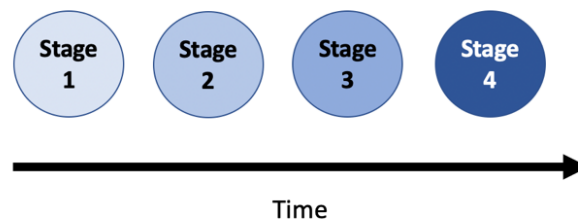


Figure 7: An example would be splitting cancer into stages 1-4

The model discussed now, called SuStaIn (Young et al, 2018), aims to combine these concepts and subtype disease *based on* progression, as is illustrated in Figure 8 below.

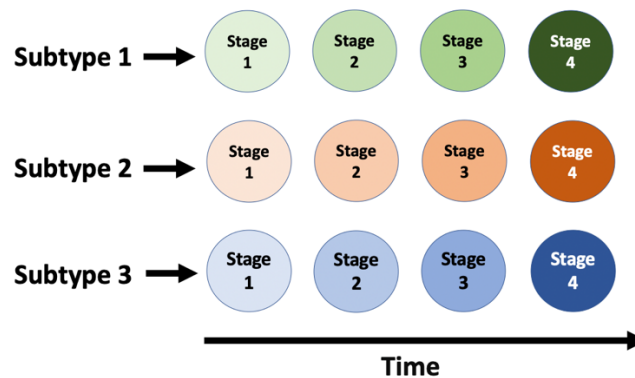


Figure 8: SuStaIn

Subtyping disease based on how it progresses

More specifically, SuStaIn is a hierarchical clustering algorithm that identifies subtypes with distinct progression patterns from cross-sectional data. Instead of inferring *either* cross-sectional subtypes or a disease progression model, it essentially does both by finding subtypes *based on* how the disease progresses.

SuStaIn learns the number of subtypes the available data can support (using cross validation), reconstructs a trajectory of stages within each subtype and assigns a probability of each subtype and stage to each subject (Young et al, 2018). It uses data likelihood (rather than distance as is used in algorithms such as K-means) to find the clusters, where likelihoods for progression patterns are provided by the Linear Z-score Model or EBM for Gaussian data, or the KDEMM for non-Gaussian data (see illustration in Figure 9 below).

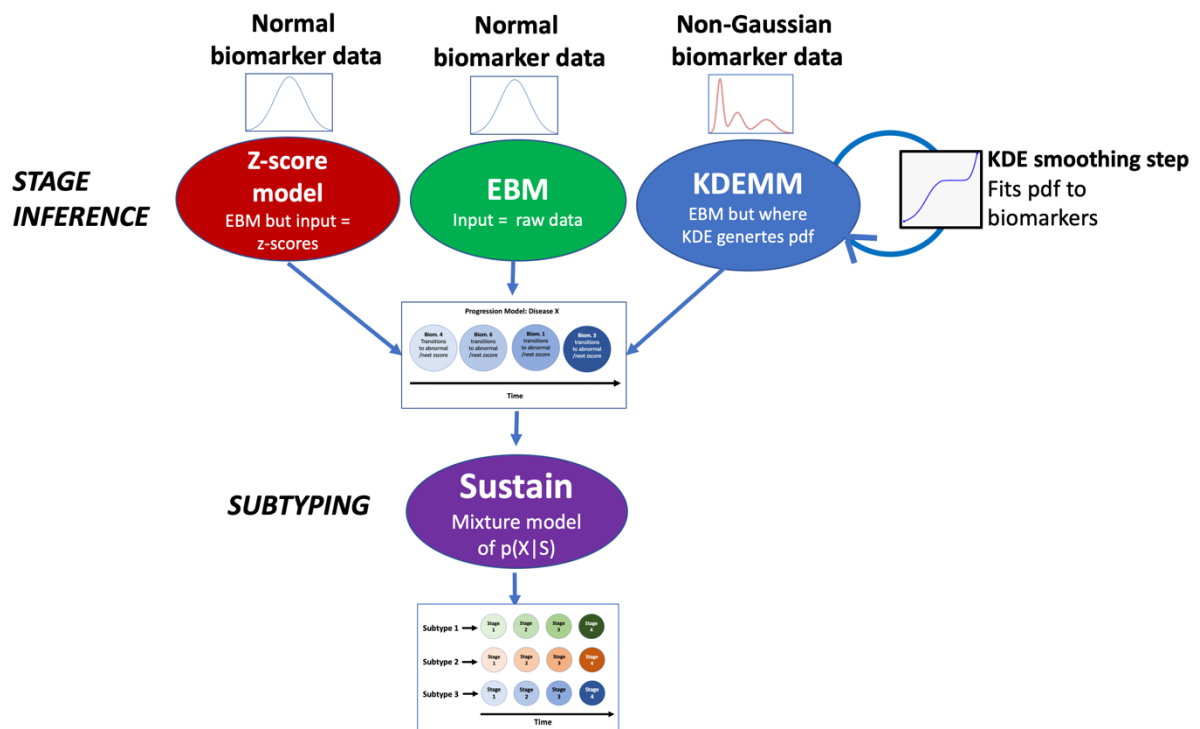


Figure 9: Sustain subtyping based on progression patterns

Based on the data type, one of the stage inference mixture models (Z-score model, EBM, KDEMM) is used to fit the stage likelihoods. The stage likelihoods are then inputted to SuStaIn which subtypes them into distinct progression patterns.

An assumption made by SuStaIn is that variance between disease features is independent. Subtypes recovered by SuStaIn have been shown to be robust to disease feature covariance, nevertheless, refinements might come from modelling covariance among strongly dependent

biomarkers. Secondly, to enable the modelling of purely cross-sectional data, an assumption is also made of a dimensionless (rather than time-based) 'timescale'.

The main advantage of SuStaIn is its unique ability to find clusters based on sequence likelihoods (i.e., to find progression-based subtypes) from cross-sectional data. This is hugely useful in medical research for the reasons mentioned previously (in section 1.8). A second advantage is that it is a highly interpretable model, something which is critical for medical applications. A drawback is that there is no measure of time in the progression models (as per the time assumption mentioned in the previous paragraph). The model gives an ordering of events, but not when they happen. However, due to the numerous nature of the progression stages (there could be 25 stages for example), this model would still likely give us a more useful progression taxonomy than the traditional progression systems (with 4-5 stages) discussed in section 1.6.1.

SuStaIn was introduced in the Young et al (2018) study where three distinct progression patterns (i.e. three subtypes) of Alzheimer's disease were found using MRI data.

1.10 Objective & hypothesis

This study will investigate whether progression-based subtypes of HD exist using clinical test score data from the Enroll-HD data set (explained in methods).

This will be done with the view to developing data-driven subtypes of HD which can be targeted during drug development. It is hypothesised that trialling drugs for specific subtypes of HD will improve the chances of developing progression-altering therapeutics for the disease.

Hypothesis 1: Progression-based subtypes of HD exist and can be quantitatively identified using clinical test score data

It is also hypothesised that these subtypes will exist independent of CAG repeat length, supporting earlier findings that symptom variation occurs even in cases with the same or

similar number of CAG repeats (discussed in section 1.5). This would support the earlier argument that clinical test score data is most useful in modelling HD progression.

Hypothesis 2: The CAG repeat distributions of each subtype will not show statistically significant differences

2 Methods

2.1 Data set

Enroll-HD is a worldwide, longitudinal (2012 - present), observational study of Huntington's disease whose overarching goal is to accelerate progress towards the development of effective therapeutics for HD (Enroll-HD Guide).

Participants in the Enroll-HD data set are individuals with either manifest HD (who carry the mutated HTT gene and display symptoms), pre-manifest HD (who carry the mutated HTT gene but do not display symptoms) or healthy controls (who do not carry the HD expansion mutation). These can be seen in Table 2 below. In this study, Gene Negative and Family Control participants were combined to form the healthy controls group.

Label	Description
Manifest/Motor-manifest HD	Carriers of the mutated HTT gene showing clinical features that are regarded by the investigator as diagnostic of HD.
Pre-Manifest/-Motor-manifest HD	Carriers of the mutated HTT gene who are <i>not</i> showing clinical features regarded as diagnostic of HD.
Genotype Negative	This group includes a first or second degree relative, i.e., related by blood to a carrier, who has undergone predictive testing for HD and is known not to carry the HD expansion mutation.
Family Control	Family members or individuals not related by blood to carriers (e.g., spouses, partners, caregivers)

Table 2: Participant groups (Enroll HD Data Guide)

Participants were asked to attend as many annual study visits as possible at sites located in North America, Latin America, Europe, Australia and New Zealand. These visits are recorded as either "Baseline" (the first visit), "Follow up" or "Phone contact".

The Enroll data set as of July 14th 2020, contained 17,187 visits from 8,714 participants, and covered 296 features across 9 data files. The data files included many feature areas including demographic, lifestyle (including cigarette, alcohol and recreational drug consumption)

medical history (including records of pharmacological and non-pharmacological therapies), and motor, cognitive, psychiatric and general functioning test scores.

Because SuStaIn requires only cross-sectional data, we only need one visit per person. As such only Baseline visits were included. Baseline was chosen to provide the maximum data volume as more patients attended the Baseline visit than any other visit type, and more information was collected on this visit than on any other visit.

With the exception of *subject ID*, *HD category* and the covariates *age* and *education level*, all non-clinical test score features were removed (83 columns) for the analysis set as they were not relevant to the study. These mainly included demographic, medical history and substance abuse data.

This data subset was then cleaned, feature-analysed (where a subset of high-signal features was selected), and adjusted for covariates (to strip out the effects of *age* and *education level* which could also affect motor or cognitive test scores). These steps are explained in the next section.

2.2 Data cleaning and normality testing

The Baseline database was cleaned using the following steps.

2.2.1 Excluding erroneous results

The variables were checked for erroneous entries such as values that were outside the scale of measurement. For example, *education level* was scaled from 0 to 6 so any values above 6 were removed. *Age* also had some entries which contained the string "<18". These were adjusted to numeric value, 17.

Additionally, the HD category variable was recoded from [2=pre-manifest, 3=disease, 4=gene negative, 5=family control] to [0=control, 1=disease, 2=pre-manifest] whereby family control and gene negative were combined.

2.2.2 Normality testing

Clinical test scores are often non-Gaussian due to floor effects. Floor effects in tests usually occur if the test is too difficult and thus, many participants get very low scores. In clinical testing, instead of the tests being too difficult, floor effects can be caused by the subjects having reduced capacity resulting from, for example, HD. Therefore it is unwise to assume clinical test scores are Gaussian.

Normality of the data was tested using the D'Agostino and Pearson's omnibus normality test which tests the null hypothesis that the data was drawn from a normal distribution. Results showed that none of the clinical test score variables in the data set were Gaussian. The non-Gaussian nature of the data was corroborated further by visual inspection of histograms of the features selected for analysis (shown later in section 2.5.4).

2.3 **Mathematical model**

The Linear Z-Score model SuStaIn was run first to see if it was possible to attain the more fine-grained staging based on z-scores. However, as expected, due to the non-Gaussian nature of the input data, poor results were returned.

Therefore, the non-parametric KDEMM SuStaIn was used to find stage-based subtypes. With this method, Kernel Density Estimation (KDE) was used to fit distributions to the disease features. The non-parametric mixture model (MM) was used to find the event likelihood and maximum likelihood sequence, and SuStaIn was used to subtype the sequence into N clusters. Each of these parts will now be explained.

2.3.1 Fitting the KDEMM

Because the KDEMM is based on the EBM it makes sense to first explain this technique.

Equation (1) shows the basic likelihood equation for the EBM.

$$P(X|S) = \prod_{i=1}^J \left[\sum_{k=0}^J \left(\prod_{i=1}^k P(x_{ij}|E_i) \prod_{i=k+1}^J P(x_{ij}|\neg E_i) \right) \right]$$

Equation (1): The Event Based Model (EBM)

$p(X|S)$ is the most likely order of events that maximises the likelihood of the data.

X is the data set of raw disease feature measurements.

S is the sequence in which disease feature turn abnormal.

$p(k)$ is the prior likelihood of being at stage k at which the events $E_1, \dots, E_k$ have occurred.

$p(x|E)$ and $p(x|\neg E)$ are the probability the event has occurred (the disease feature has transitioned to abnormal) and not occurred. These are modelled as normal distributions.

The model uses a uniform prior on the stage so that a priori individuals are equally likely to belong to any stage along the progression pattern.

When only a small number of disease features is used, all possible orderings can be enumerated and the characteristic ordering, $\hat{S}$, is the sequence which maximises $p(X|S)$. However, as the number of possible sequences grows factorially as the number of disease feature increases, Markov Chain Monte Carlo (MCMC) is used to find $\hat{S}$ when the number of sequences to sample is too large. (MCMC is a technique that allows us to estimate the shape of a posterior distribution in cases where we can't compute it directly. It does so by approximating the posterior distribution of a parameter of interest by random sampling in a probabilistic space).

Once we have $\hat{S}$, it can be used to estimate a disease stage for individuals given their disease feature measurements. The disease stage, k , is defined as, the stage (i.e. the number of events that have occurred so far) that has the highest probability given the data and our sequence, as seen in Equation (2), (Firth et al, 2018).

$$\operatorname{argmax}_k P(X_i|S, k) \prod_{i=1}^k P(x_{ij}|E_i) \prod_{i=k+1}^J P(x_{ij}|\neg E_i)$$

Equation (2): Maximizing EBM (Firth et al, 2018).

In KDEMM we do the same thing, except $p(x|E)$ and $p(x|\neg E)$ in Equation (1) are now given by the KDE model in Equation (3) below (instead of by a Gaussian Mixture Model).

$$\hat{f}(x) = \frac{1}{nh} \sum_i K\left(\frac{x - x_i}{h}\right)$$

Equation (3): KDE (Firth et al, 2018).

The ordering, S , is then found using Expectation Maximisation (using the likelihood function from the KDE) which, similar to the k-means algorithm, iterates over alternating assignment and updates steps to optimize parameters (Firth et al, 2018). Equation (4) shows the update step whereby for each subset of the data $S_j^{(t)}$ a component $f_{S_j^{(t)}}$ is fit (Firth et al, 2018).

$$\hat{f}_{S_j^{(t)}}(x_i) = \frac{m_j^{(t)}}{nh} \sum_{x_j \in S_j} K\left(\frac{x_i - x_j}{h}\right)$$

Equation (4): Update Step, KDEMM (Firth et al, 2018).

In the assignment step each observation is then assigned to a new subset, $S_j^{(t+1)}$, to which it has the maximum likelihood of belonging, i.e. $f_{S_j^{(t)}}(x_i)$. Mixture weights are then updated to be the proportion of observations in each subset as in Equation (5).

$$m_j^{(t+1)} = |S_j^{(t+1)}| / \sum_{i=1}^k |S_i^{(t+1)}|$$

Equation (5): Mixture weight updating (Firth et al, 2018).

The update and assignment steps are then iterated until subset assignment is no longer updated.

2.3.2 Fitting SuStaIn

SuStaIn model fitting requires simultaneously optimising subtype membership, subtype trajectory and the posterior distributions of both (Young et al, 2018).

$$P(\mathbf{X}|\mathbf{M}) = \sum_{c=1}^C f_c P(\mathbf{X}|\mathbf{S}_c)$$

Equation (6): The SuStaIn model

C is the number of clusters (subtypes)

f is the proportion of subjects assigned to a particular cluster (subtype)

$\mathbf{M}$ is the overall SuStaIn model

The SuStaIn model is fitted hierarchically, whereby the number of subtypes that can be supported by the data is selected via model selection criteria CVIC (cross-validation information criterion; Gelman et al, 2014) which is obtained from ten-fold cross-validation. The hierarchical fitting initialises the fitting of each C -cluster (subtype) model from the previous $C-1$ -cluster model, i.e. the clustering problem is solved sequentially from $C = 1 \dots C_{\max}$ (where $C_{\max}$ is the maximum number of clusters being fitted), initialising each model using the previous model. For the initial cluster ($C = 1$), we use the single-cluster expectation maximisation (E-M) procedure (explained in Appendix 3). We fit subsequent cluster numbers ($C > 1$) hierarchically by generating $C-1$ candidate C -cluster models using a split-cluster E-M procedure (also explained in Appendix 3). From these $C-1$ candidate C -cluster models, the model with the highest likelihood is chosen (Young et al, 2018).

2.3.3 Uncertainty estimation (MCMC)

In addition to estimating the most probable sequence, S , for each subtype, we can determine the relative likelihood of all sequences for each subtype by evaluating the probability of each possible sequence. This gives us an estimate of the uncertainty in the ordering S (Young et al, 2018). In practice the number of sequences is too large to evaluate all possible sequences so we use Markov Chain Monte Carlo (MCMC) sampling is used to provide an approximation of the uncertainty.

In this study we take 100,000 MCMC samples initiated from the maximum likelihood solution. We will later check that the MCMC trace shows good mixing properties.

2.4 Feature analysis

To select the best set of data features for use in this model, Wilcoxon test and random forest feature selection were used, both of which are non-parametric and suitable for non-Gaussian data. This was done on the cleaned data set containing 8,714 participants' Baseline visits.

2.4.1 Wilcoxon-test feature analysis

The Wilcoxon signed-rank test tests the null hypothesis that two related paired samples come from the same distribution. It is essentially a non-parametric version of the paired T-test which helps us identify which features have the highest effect size. This was run across all 213 clinical test-score features. The 15 features with highest effect sizes and a p-value of <0.05 are listed below.

Variable Name	Code	Category	Statistic	P-value
Depression score (number)	depscore	Psychiatric	476674.0	<0.01
Anxiety, Frequency (0-4)	pbas3fr	Psychiatric	380936.0	<0.01
Anxiety, Severity (0-4)	pbas3sv	Psychiatric	343319.5	<0.01
Depression, Frequency (0-4)	pbas1fr	Psychiatric	265399.0	<0.01
Irritability and aggression (0-4)	irascore	Psychiatric	250124.0	<0.01
Irritability, Frequency (0-4)	pbas4fr	Psychiatric	237868.0	<0.01
Anxiety, Severity (0-4)	pbas4sv	Psychiatric	211161.0	<0.01
Verbal fluency test (category), total pervasive errors (number)	verfct7	Cognitive	211061.5	<0.01
Symbol Digit Modality Test, total errors (number)	sdmt2	Cognitive	169038.5	<0.01
Mental component (number)	mcs	Psychiatric	167605.5	<0.01
Verbal fluency test (letters), total pervasive errors (number)	verflt07	Cognitive	160405.0	<0.01
Stroop inference test, self-corrected errors (number)	sit3	Cognitive	151156.5	<0.01
Stroop colour naming test, self-corrected errors (number)	sent3	Cognitive	144516.5	<0.01
Anxiety subscore (number)	anxscore	Psychiatric	125526.0	<0.01
Irritability subscore	irrscore	Psychiatric	103380.5	<0.01

Table 3: Results of Wilcoxon test feature analysis

2.4.2 Random forest feature analysis

Random forest (RF) is a supervised non-parametric ensemble learning technique consisting of the aggregation of a large number of decision trees, resulting in a reduction of variance compared to the single decision trees (Couronne et al, 2019). RF also has the advantage of being able to rank the variable importance measures (VIMs) of the features, with respect to their relevance in predicting the outcome (Couronne et al, 2019).

In the Enroll-HD data, the feature "hdcat" (HD category) was used as the target/outcome variable. The top 15 results are shown in Table 4.

Variable Name	Code	Category	VIM
Total motor score, incomplete (number)	miscore	Motor	0.12
Chorea, RUE (0-4)	chorrue	Motor	0.07
Saccade initiation horizontal (0-4)	sacinith	Motor	0.05
Saccade velocity vertical (0-4)	sacvelv	Motor	0.05
Independence (%)	indep scl	Functional	0.05
Finger taps, right (0-4)	fingtapr	Motor	0.05
Finger taps, left (0-4)	fingtapl	Motor	0.04
Total functional capacity (number)	tfcscore	Functional	0.04
Chorea, BOL (0-4)	chorbol	Motor	0.04
Chorea, LUE (0-4)	chorlue	Motor	0.03
Chorea, LLE (0-4)	chorlle	Motor	0.03
Could engage in employment (0/1)	emplusl	Functional	0.03
Total motor score (number)	motscore	Motor	0.03
Ocular pursuit, horizontal (0-4)	ocularh	Motor	0.02
Saccade initiation, vertical (0-4)	sacinitv	Motor	0.02

Table 4: Random Forest Variable Importance Measure (VIM) results

To maximise the chance of finding subtypes based on different types of progression, a key objective was to select variables from across the symptom areas of motor, cognitive and psychiatric decline. A second objective was to select features with high signal (using Wilcoxon and RF), and a third was to select features for which a good fit could be found using KDE. A balance of these three objectives was used to choose the final features for analysis which are shown in Table 5.

Variable Name	Code	Category
Total motor score (number)	motscore	Motor
Chorea, RUE (0-4)	chorrue	Motor
Saccade initiation horizontal (0-4)	sacinith	Motor
Symbol Digit Modality Test, total errors (number)	sdmt2	Cognitive
Verbal fluency test (category), total pervasive errors (number)	verfct7	Cognitive
Trailmaking test, total errors (number)	trlb3	Cognitive
Depression score (number)	depscore	Psychiatric
Irritability, aggression (number)	irascore	Psychiatric
Anxiety, severity (number)	pbas3sv	Psychiatric

Table 5: Feature selected for analysis

2.5 Data cleaning: Part 2

2.5.1 Handling missing values

Two approaches were considered for dealing with missing values in the new subset; (1) dropping rows containing a NaN value, and (2) replacing NaN values with the column mean ('mean imputation'). Dropping is often preferred as it avoids introducing any bias via a sample mean (which can be problematic if the data is itself is biased, for example on small data sets with large variance). However, if the data set contains a lot of NaN values, we could end up dramatically reducing the size of our data set, and as such, our statistical power.

In this data subset, NaN values totalled 30.5%. This was considered sufficiently small to use the dropping approach in order to avoid the risk of biasing the data with mean imputation. A data set of 6,206 rows remained.

2.5.2 Handling outliers

Data points which were more than 5 standard deviations from the mean were counted. Outliers of this nature only represented around 5% of the dataset, and so were removed.

2.5.3 Adjusting for covariates

There are several factors which could influence motor, cognitive, and psychiatric test outcomes other than Huntington's disease. For example, cognitive and motor ability tends to decline with age after a certain point. Additionally, education level can positively influence cognitive test scores.

We want to minimize the effect of these variables on the results so that we only see the disease effects. Therefore before the data was analysed, the effects of these covariates were regressed out. This was done on the variables *education level* ('*isced*') and *age* ('*age*').

2.5.4 Data subset used in analysis

The final data set used after feature selection and data cleaning contained 5,947 participants, the demographic breakdown of which are below (all figures rounded to integer).

Disease/control: Disease = 47%, Control = 28%, Pre-manifest = 25%

Sex: Females = 56%, Males = 44%

Region: Europe = 52%, Northern America = 43%, Australasia = 4%, Latin America = 1%

Race: Caucasian = 92%, Hispanic or Latino = 2%, American Black = 1%, Native American = 1%, Asian = 1%, Mixed = 1%, Other = 2%

The feature distributions are shown in Figure 10 below. As expected (and discussed in section 1.8.1 and 2.2.2), they are not Gaussian-distributed. This illustrates the need for a non-parametric model and confirms the decision to use KDEMM SuStaIn for subtyping the data.

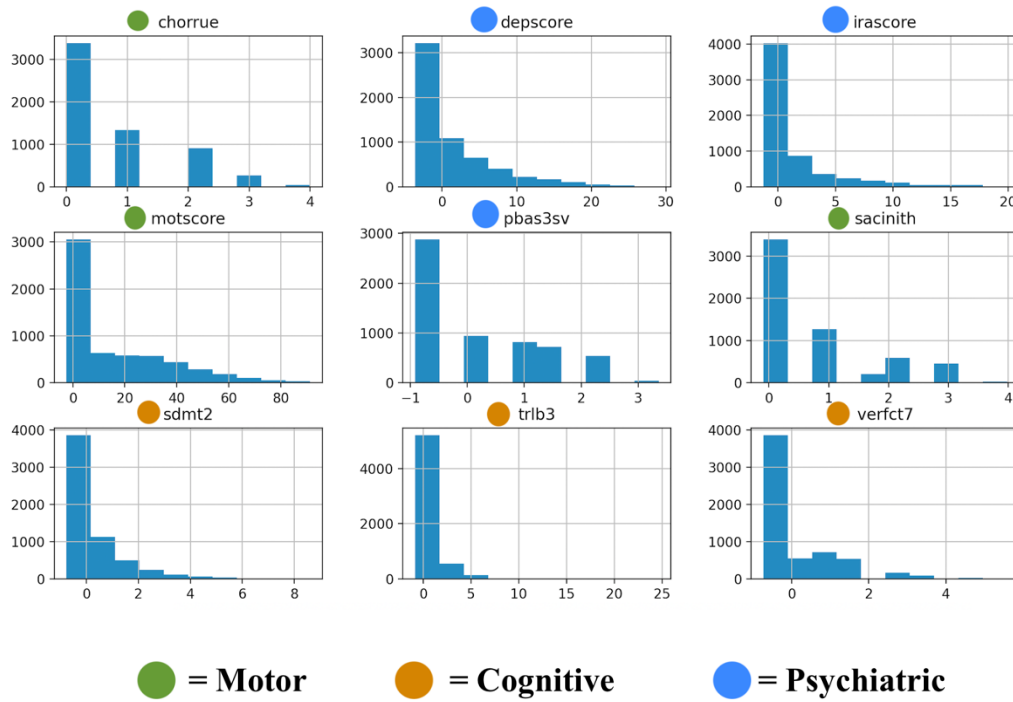


Figure 10: Distributions of the 9 disease features used in analysis

3 Results

3.1.1 KDE fits

Before running SuStaIn, the non-parametric smoothing function, KDE (Kernel Density Estimation), was run on all 9 variables in order to fit distributions.

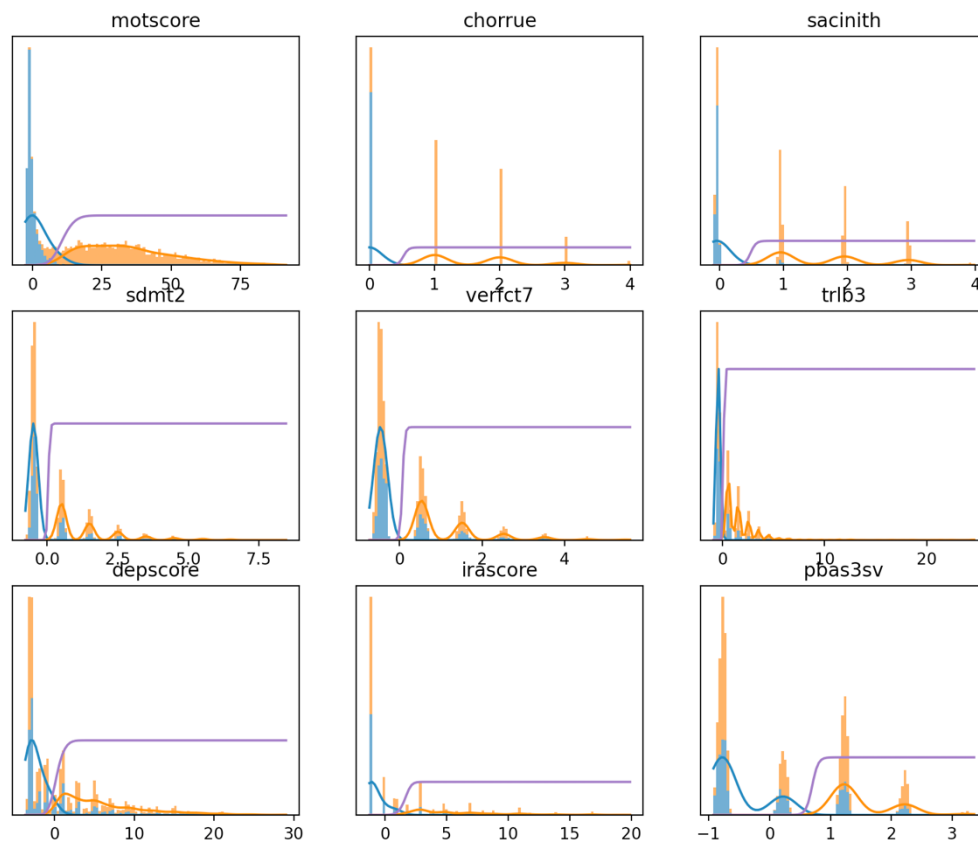


Figure 11: KDE-fitted distributions for all 9 features

Orange shows disease cases. Blue shows control cases. The purple line shows probability of disease. As shown, a good KDE fit was found for all features which is indicated by the monotonic increase in all purple lines.

(Top row = Motor scores. Middle row = Cognitive scores. Bottom row = Psychiatric scores).

The KDEMM SuStaIn was then run which outputs several types of result; positional variance diagrams, cross validation diagrams and MCMC likelihoods traces. These are shown below.

3.1.2 Positional variance diagrams

Positional variance diagrams show us (1) the order in which biomarkers turn abnormal in each subtype, (2) the proportion of subjects, f , that fall into each subtype and (3) the number of participants, n , that fall into each subtype.

The positional variance diagram for the analysis can be seen below in Figure 12 below.

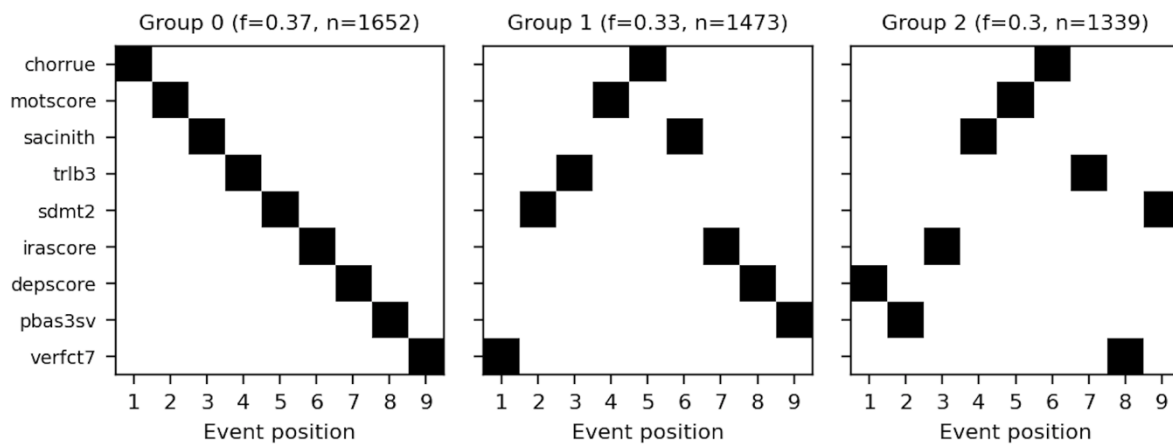


Figure 12: Positional Variance Diagrams

The **y-axis** shows the biomarkers. The **x-axis** shows the event ordering. **Group** refers to the subtype. **f** represents the proportion of subjects in that subtype. **n** represents the number of participants in the group.

Motor tests = chorrue, motscore, sacinith

Cognitive tests = trlb3, sdmt2, verfct7

Psychiatric tests = irascore, depscore, pbas3sv

3.1.3 Cross validation

The number of subtypes that can be supported by the data is selected via model selection criteria CVIC which is obtained from ten-fold cross-validation.

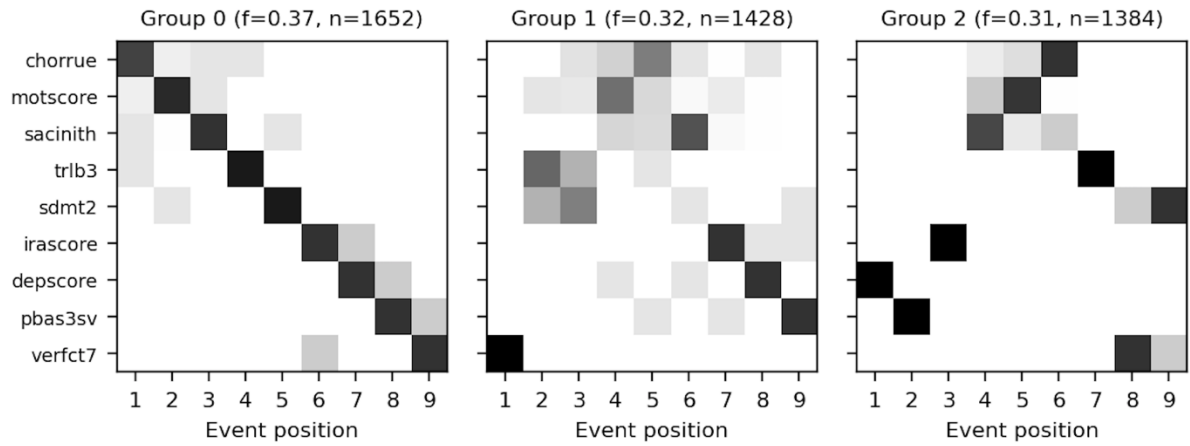


Figure 13: Cross validation diagrams for the three subtypes

The darker squares indicate more certainty; the lighter squares indicate uncertainty.

3.1.4 MCMC subtype likelihoods

As discussed earlier, Markov Chain Monte Carlo was used to determine the relative likelihood of all sequences for each subtype. This was done by evaluating the probability of each possible sequence.

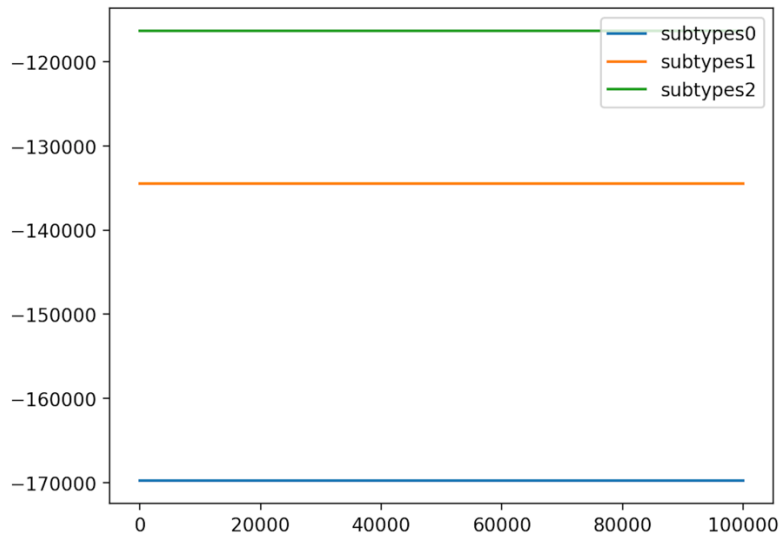


Figure 14: MCMC Likelihood traces

y axis = log likelihood. **x axis** = iterations up to 100,000 (set by the researcher).

Subtype 0 ('1') = [-16,985.59], Subtype 1 ('2') = [-13,448.53], Subtype 2 ('3') = [-11,704.5]

3.1.5 T-tests between subtypes

To test if there was a significant difference between subtypes, t-tests were run between subtypes 1 & 2 and between subtypes 2 & 3. The results can be seen in Table 6 below.

	T-statistic	P-value
Subtypes 1 & 2	-21.99	1.85166581e-14
Subtypes 2 & 3	-13.24	1.01673757e-10

Tables 6: T-tests between subtypes

3.1.6 T-tests between the CAG distribution of each subtype

To test if variance in symptom progression existed independently of CAG repeat length, t-tests were done between the CAG distribution of each subtype.

	T-statistic	P-value
Subtypes 1 & 2	0.2836	0.7766
Subtypes 2 & 3	-0.8949	0.3709
Subtypes 1 & 3	-0.7343	0.4628

Tables 7: T-tests between the CAG distributions of the subtypes

4 Discussion

4.1.1 General discussion of results

The results will first be discussed in relation to hypothesis 1 and then in relation to hypothesis 2.

Hypothesis 1: *Progression-based subtypes of HD exist and can be quantitatively identified using clinical test score data*

The positional variance diagrams (PVDs) show three subtypes with distinct progression patterns. Group 1 shows a subtype where motor decline occurs first (*chorrue*, *motscore* and *sacinith* are motor tests), followed by cognitive decline (*trlb*, *sdmt2*) followed by psychiatric decline (*irascore*, *depscore*, *pbas3sy*; irritability, depression and anxiety). Group 2 shows a subtype where cognitive decline clearly happens first (*trlb3*, *sdmt2*, *verfct7*), followed by motor then psychiatric decline. Group 3 shows a pattern where psychiatric symptoms come first followed by motor then cognitive. These results are illustrated in Figure 15 below.

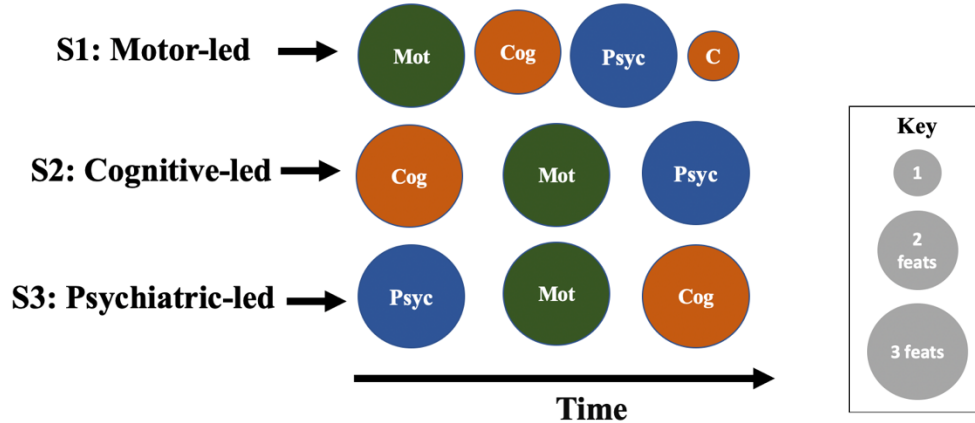


Figure 15: Subtypes illustration

The largest circle indicates 3/3 features from the labelled symptom category. Middle sized circle = 2/3 subtypes from the category. Small circle = 1/3 subtypes from the category. *feats* = features.

The t-test results between S1 and S2, and between S2 and S3 supported the statistical significance of these distinct subtypes. The MCMC trace (Figure 14) shows good mixing as the likelihoods of the subtypes remains flat from 1 up to 100,000 iterations (poor mixing would exhibit "wavy" or snake-like lines).

The cross validation diagrams (Figure 13) indicate that there is little uncertainty in Subtype 1 (Motor-led), shown by the strong dark squares in the same pattern as on the corresponding PVD diagram. Subtype 2 has the most uncertainty, especially around the early and middle stages. However, it shows strong certainty that psychiatric decline occurs last. Subtype 3 shows no uncertainty around psychiatric decline occurring first, but shows a little uncertainty about the order of the motor features in the middle stages, and the order of psychiatric features in the latter stage.

Overall, whilst there is some uncertainty in the model, there is strong evidence to support the findings that in Subtype 1, motor decline occurs first, in Subtype 3, psychiatric decline occurs first and in Subtype 2, psychiatric decline occurs last. It ultimately supports that there are three distinct subtypes.

Another interesting observation from the PVDs is that the proportions of subjects in each category are reasonably similar with 37% motor-led, 33% cognitive-led, and 30% psychiatric-led. This indicates that there is no largely dominant subtype of the disease.

As would be expected, the onset of symptoms within a category (motor, cognitive or psychiatric decline) tend to occur together. Looking at the features used in this study, that means for example, *anxiety*, *depression* and *irritability/aggression* are likely to occur together and *chorea*, *eye movement disfunction* and *overall motor score* are also likely to occur together.

Hypothesis 2: *The CAG repeat distributions of each subtype will not show statistically significant differences.*

The t-tests results between CAG distributions did not show significant differences, strongly indicating that our distinct progression patterns (subtypes) exist independently of CAG repeat length. This fortifies earlier findings (by Georgiou et al, 1999; Friedman et al, 2005; Gome-Esteban et al, 2007; Thu et al, 2010) that symptomatic variance exists independently of CAG repeat. It also supports the argument made in this paper, that whilst CAG repeat length may be useful in determining *whether* someone will show symptoms, it is not a suitable biomarker for predicting *types* of symptoms, or symptom-based disease progression.

In summary, both hypotheses in this paper were supported by the results.

4.1.2 Medical relevance

As noted in the literature review, the well established Diagnostic & Statistical Manual of Mental Disorders (DSM IV), states that "the onset of motor abnormalities tends to occur in the third or fourth decades of life, however cognitive and psychiatric symptoms can occur much earlier than this". Perhaps one of the most interesting and controversial things about the findings in this paper is that in the most dominant subtype, *Motor-led HD*, contradicts this; it shows that motor symptoms come *before* cognitive and psychiatric decline in this subtype. This highlights a potential need to redefine HD in this clinical handbook used by many psychologists and researchers.

As discussed throughout this study, the identification of subtypes could lead to improved patient stratification in drug trials. To date, no drug has been approved which can alter the progression of HD. The subtypes shown in Figure 15 could be a clue as to why. The findings in this study indicate that clinical trials so far have attempted to alter 3 different progression patterns with one drug. If going forward, cohorts in drug trials are split into *Motor-led HD*, *Cognitive-led HD*, *Psychiatric-led HD* and *Controls*, we may begin to see that certain drugs start working on certain types of the illness.

Stratifying patients in drug trials by these subtypes could also help ensure the validity of results. It could prevent researchers from incorrectly attributing a slowing of symptom decline to drugs, when the slowing is actually down to the natural pattern of the subtype. For example in *Motor-led HD* the majority of motor decline passes at the start, therefore further decline is naturally unlikely after a certain point. It would be incorrect to attribute that to a drug.

In addition to drug discovery, identification of the subtypes shown in Figure 15 could also have a big impact on medical treatment and support for people who are trying to manage HD symptoms. For example we can see that people with *Psychiatric-led HD* have a marked psychiatric decline early on. Knowing an early psychiatric decline is likely to occur could allow doctors to allocate psychiatric support to patients early enough to be effective. Likewise, knowing that significant cognitive decline in *Psychiatric-led HD* patients may not

occur until the very late stages, could help patients themselves plan to continue working for longer (if needed), for example by arranging a working-from-home set-up with some physical support. Similarly, for *Motor-led HD* patients, physical support is clearly going to be needed much earlier on. And for *Cognitive-led HD* patients, cognitive support will be needed first.

This subtyping and staging system could also be used to more accurately forecast requirements of HD patients on a health board level. This could lead to better treatment allocation and helping ensure people get the care they need. For example, instead of splitting a psychologist's working week equally between three HD patients (one each from *Motor-led*, *Cognitive-led* and *Psychiatric-led*), the psychologist could instead start by spending the majority of their time with the *Psychiatric-led HD* patient. The following year (or whichever time period is chosen), the psychologist can start to rebalance their time to spend more with the *Motor-led HD* patient. Then more time with the *Cognitive-led* patient the year after that, when their psychiatric decline is expected.

One of the major drawbacks of SuStaIn is that it does not currently give us more than an arbitrary timeline for progression. But in defining these subtypes, we are providing a foundation upon which real timeline could be built (a suggestion for this is given in the next section).

4.1.3 Future work

Within the Enroll-HD dataset, a useful next step would be to test whether each of these groups responded differently to different treatments, i.e., if clinical outcomes following a treatment differ between subgroups (Enroll-HD includes a data set on administered treatments). Given the treatments for HD are prescribed based on symptoms, the results of this test will have limited bearing on whether the same applies to treatments aimed at altering the course of HD. However, showing that subtypes respond differently to treatments strengthens the case that clinical subtypes exist and could encourage the stratification of patients in drug trial cohorts.

Looking beyond the Enroll-HD data set, the most interesting next step for this study would be to use these subtypes to stratify patients in the current clinical trials for gene-altering therapy

RG6042 (by Tabrizi et al 2019), and to monitor whether each group responds differently to the drug.

A very useful development, which could be built on the foundation of this paper, would be to monitor patients within these subtypes over time, and measure the amount of time they spend in each stage. This could provide an actual (rather than dimensionless) time scale to the subtypes.

Lastly, the subtypes identified may also link back to distinct patterns of neuro-decay. It could be hypothesised for example, that in *Psychiatric-led HD*, neurodegeneration starts in areas responsible for dopamine projection, such as the globus pallidus and substantia nigra in the basal ganglia. In *Motor-led HD*, neurodegeneration may be more likely to start in the striatum and motor cortex. Therefore it would be interesting to run the KDEMM SuStaIn on MRI brain images of people with HD to see if neurodegeneration follows patterns that reflect the *Motor-led*, *Cognitive-led* and *Psychiatric-led* patterns found in this study.

4.1.4 Conclusion

In conclusion, this research has identified three statistically different subtypes of Huntington's Disease, each with distinct symptomatic progression patterns. This is the first time such subtypes have been found. These subtypes can be defined as *Motor-led HD (Subtype 1)*, *Cognitive-led HD (Subtype 2)* and *Psychiatric-led HD (Subtype 3)*. In addition, a more fine-grained, data-driven staging system was identified for each subtype. This is an improvement on traditional, subjectively developed, staging systems currently in use.

By treating Huntington's as three separate sub-diseases, and following the staging pattern of each, treatment could be planned in a way that improves patient outcomes and potentially even health board resource planning.

Moreover, clinical trial cohorts could be split by subtype, which could allow the discovery of precision drugs that work on certain subtypes. Ultimately, treating HD as separate sub-diseases in drug trials, could improve the chance of finally finding a progression-altering cure.

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6 Appendices

6.1.1 Appendix 1: Vonsattel Grading System for HD (taken from Bates et al, 2014)

Grade 0 comprises less than 1% of HD brains. Gross examination shows features indistinguishable from normal brains. On general survey using LHE- or HE-stained slides alone, neither reactive gliosis nor neuronal loss is reliably detectable. However, further evaluations including cell counts indicate a 30% to 40% loss of neurons in the HCN and no visible reactive astrocytosis. Furthermore, a study using immunohistochemistry and sections of three presymptomatic gene carriers revealed ubiquitinated, nuclear inclusions in all three brains, including one individual, with 37 polyQ, who died putatively three decades before the expected age for onset of symptoms (Gomez-Tortosa et al., 2001). In addition, cell counts of the TCN revealed an increased density of oligodendrocytes

In **Grade 4**, the striatum is severely atrophic (the medial contour of the HCN is concave, as is the caudate nucleus at the anterior limb of internal capsule). The neostriatum loss is 95% or more neurons. In at least 50% of grade 4 brains, the underlying nucleus accumbens remains relatively preserved but is not normal.

among the presymptomatic HD gene carriers (Gomez-Tortosa et al., 2001).

Grade 1 comprises 4% of HD brains.

The TCN is smaller than normal, as most likely is the BCN. Neuronal loss and astroglia involve the TCN, BCN, and dorsal portion of both the head and nearby dorsal putamen. Cell counts show 50% or greater loss of neurons in the HCN.

Grade 2 comprise 16%, those assigned **grade 3** comprise 52%, and those assigned **grade 4** comprise 28% of all HD brains. Gross striatal atrophy is mild to moderate in grade 2 (the medial outline of the HCN is only slightly convex but still bulges into the lateral ventricle) and severe in grade 3 (the medial outline of the HCN forms a straight line or is slightly concave medially). Thus, the microscopical changes in grades 2 and grade 3 are more severe than those in grade 1 and less than in grade 4 brains.

6.1.2 Appendix 2: Shoulson & Fahn (1979) Stage Classification of HD

	Engagement in occupation	Capacity to handle financial affairs	Capacity to manage domestic responsibilities	Capacity to perform activities of daily living	Care can be provided at
Stage 1	Usual level	Full	Full	Full	Home
Stage 2	Lower level	Requires slight assistance	Full	Full	Home
Stage 3	Marginal	Requires major assistance	Impaired	Mildly impaired	Home
Stage 4	Unable	Unable	Unable	Moderately impaired	Home or extended care facility
Stage 5	Unable	Unable	Unable	Severely impaired	Total care facility only

6.1.3 Appendix 3: Single Cluster E-M Procedure & Split Cluster E-M procedures

Referenced in the SuStaIn explanation in section 2.3.2. Explanations taken from Young et al (2018).

Single Cluster E-M Procedure

The single-cluster E-M procedure is used to find the optimal model parameters (the sequence S in which the biomarkers reach each z-score) for a single-cluster. In the single-cluster E-M procedure the sequence S is initialised randomly. This sequence is then optimised using E-M by going through each z-score event E in turn and finding its optimal position in the sequence relative to the other z-score events, i.e. by fixing the order of the subsequence $T = S/E$ and maximising the likelihood of the sequence by changing the position of event e in the subsequence T . The sequence S is updated until convergence. Again the single-cluster sequence S is optimised from 25 different start points (random cluster assignments) to find the maximum likelihood solution (see Convergence).

Split Cluster E-M Procedure

The split-cluster E-M procedure is used to generate each of the $C-1$ candidate C cluster models. For each of the $C-1$ clusters, the split-cluster E-M procedure first finds the optimal split of cluster c into two clusters. To find the optimal split of cluster c into two clusters, the data points belonging to cluster c are randomly assigned to two separate clusters. The optimal model parameters for these two data subsets are then obtained using the single-cluster E-M procedure. These cluster parameters are used to initialise the fitting of a two-cluster model to the subset of the data belonging to cluster c , using E-M. This two-cluster solution is then used together with the other $C-2$ clusters to initialise the fitting of the C -cluster model. The C -cluster model is then optimised using E-M, alternating between updating the sequences S_c for each cluster and the fractions f_c . This procedure is repeated for 25 different start points (random cluster assignments) to find the maximum likelihood solution.

6.1.4 Appendix 4: Pre-processing python code

```
import pandas as pd
import numpy as np
from sklearn import linear_model
import scipy.stats

#####

# CONTENTS
# (1) Fix erroneous values & Re-code hcat
# (2) Delete medical history / demographic features (leaving only test-related features)
# (3) Missing values functions
# (4) Remove outliers function
# (5) Normality test
# (6) Regress out covariates

#####

# ----- (1) FIX ERRONEOUS VALUES / RE-CODE HCAT -----

def fix_erroneous(df):
    # Replace <18 ("age") with 17 (making it numeric)
    df.replace("<18", 17, inplace=True) # Replaces <18 in "age" with 17 (making it numeric)

    # Remove isceds > 6 (scale ends at 6)
    df['isced'].value_counts() # Count values
    data = df.drop(df[df.isced > 6].index) # isced is scored 0-6

    return data

def re_code_hcat(df):
    # Re-code hcat values: 0=control, 1=manifest, 2=pre-manifest
    df["hcat"].replace(3, 1, inplace=True)
    df["hcat"].replace(4, 0, inplace=True)
    df["hcat"].replace(5, 0, inplace=True)
    return df

# ---- (2) DELETE MEDICAL HISTORY AND DEMOGRTAPHIIC FEATURES (TO GET CLINICAL TESTS ONLY) ----

def delete_med_dem(df): # Removes all demographic, medical history and drug abuse variables
    (for feature selection)
    final_col = df.columns.get_loc("updhdh")
    columns = list(df.iloc[:, 1:final_col+1].columns)
    hcat, age, isced = df["hcat"], df["age"], df["isced"]
    # print("columns to drop =", columns)
    df.drop(axis=1, labels=columns, inplace=True)
    data = pd.concat([df, hcat, age, isced], axis=1)
    return data # Returns only clinical test features

# ----- (3) MISSING VALUES -----

def missing_vals(df, print_miss_counts, method):
    if print_miss_counts == True: # This prints number of missing values in each feature
        pd.set_option('display.max_rows', None)
        print(df.isnull().sum())
    if method == "drop": # Drop missing values
        data = df.dropna()
    elif method == "mean": # Replace missing value with col mean
        disease = df[df["hcat"]==1]
        control = df[df["hcat"]==0]
        pre_man = df[df["hcat"]==2]
        # print("disease.mean() =", disease.mean())
        disease = disease.fillna(disease.mean())
        control = control.fillna(control.mean())
        pre_man = pre_man.fillna(pre_man.mean())
        data = pd.concat([disease, control, pre_man], axis=0)
    else:
        print("method must be either 'mean' or 'drop'")
    return data
```

```

# ----- (4) REMOVING OUTLIERS -----

# > Counts outliers more than 5 (can be set) SDs above or below the mean
def outlier_removal(data, features_list, sd_num=5, remove=False):
    for i in features_list[0:-1]:
        print(i)
        rows1 = data.shape[0] # For working out total outliers per column
        sd_col = data[i].std()
        print("sd =", sd_col)
        data.drop(data[data[i] > sd_num * sd_col].index, inplace=True, axis=0)
        data.drop(data[data[i] < -sd_num * sd_col].index, inplace=True, axis=0)
        rows2 = data.shape[0] # For working out total outliers per column
        outlier_count = rows1-rows2
        print("outlier_count =", outlier_count)
    return data

# ----- (5) NORMALITY TEST -----

def normality_test(data):
    for column in data.iloc[:, 1:]: # skipping subjid
        title = column
        column = data[column]
        # print(column.head)
        # print("column.shape", column.shape)
        # print("column.mean", column.mean)
        stat, p = scipy.stats.normaltest(column)
        # if p < 0.05:
        print(title, round(stat, 4), round(p, 8))

# ----- (6) REGRESSING OUT COVARIATES FUNCTIONS -----

# > Regresses out covariates
# > Prepares return data:
# - If sustainType='mixture_GMM' or "mixture_KDE" - Adds labels column back on before
  returning (KDE uses labels col) - returns hcat = 0,1,2 (controls, disease, pre-man)
# - If sustainType='ZScoreSustain' - (1) Calculates means and SDs (2) Gets Z scores (3)
  Adjusts biomarkers that decrease with disease progression - returns disease only (hcat=1?)

# Code for this function by Dr Peter Wijeratne:
def nu_freq(X_reg, y_reg): # Get regression params for each biomarker
    lr = linear_model.LinearRegression()
    lr.fit(X_reg, y_reg)
    return lr.coef_, lr.intercept_

# Code for this function by Dr Peter Wijeratne:
def controls_residuals(X, y, covar): # y used only to separate controls from disease group
    controls_covar = covar[y == 0] # taking covariates where y are controls
    controls_X = X[y == 0] # taking predictor data where y are controls
    n_biomarkers = controls_X.shape[1] # Number of biomarkers = num of columns in predictor
    data
    regr_params = dict((x, None) for x in range(n_biomarkers)) # {0: None, 1: None, 2: None,
    3: None, 4: None} - i.e., creates a key in a dict for each biomarker, with value
None:
    for i in range(n_biomarkers): # Loop creates regression coef for each biomarker
        mask = ~np.isnan(controls_X[:, i]) & ~np.isnan(controls_covar).any(axis=1)
        # applying isnan() to all rows in biomarker i |&| applying isnan() to all covariates.
        # tilde = returns complement
        X_reg, y_reg = controls_covar[mask, :], controls_X[mask, i] # ??? Applying(?) mask to
        the covariates and predictor data - putting in X_reg, y_reg
        # Could line below be done directly on controls_covar and controls_X if NaNs have
        already been removed?
        regr_params[i] = nu_freq(X_reg, y_reg) # Get regression coef and intercept for each
        biomarker and put them in values of dict
    residuals = X.copy() # Just creating a shallow copy of X
    for key, value in regr_params.items(): # TAKING THE COEFFICIENTS AWAY FROM EACH VALUE IN
        EACH BIOMARKER. For each biomarker in the regr_params dict
        residuals[:, key] -= regr_params[key][1] # resids[] = resids[] - regr_params[]
        residuals[:, key] -= np.matmul(covar, regr_params[key][0]) # resids[] = resids[] -
        regr_params[]
    # data = np.round((residuals), 4) # Round all values to 4 decimal places
    return residuals # Produces adjusted residuals

```

```

def covariate_adjustment(data_w_covariates, sustainType, num_predictors, return_stats=False):
    # (a) Getting input data for controls_residuals()
    # array_of_data_to_be_adjusted.shape = (Number of people, Number of biomarkers)
    array_of_covariate_data = np.array(data_w_covariates[["age", "isced"]], dtype=np.float64)
    # Selecting only the covariates
    # array_of_clinical_labels.shape = (Number of people)
    array_of_clinical_labels = np.array(data_w_covariates["hdcat"]) # Selecting only the
    label column
    # array_of_covariate_data.shape = (Number of people, Number of covariates)
    array_of_data_to_be_adjusted = np.array(data_w_covariates.iloc[:, 0:num_predictors],
    dtype=np.float64) # Selecting only the predictor columns

    # (b) RUN controls residuals:
    data_adjusted = controls_residuals(array_of_data_to_be_adjusted,
    array_of_clinical_labels, array_of_covariate_data)

    # (c) Prepare data for return depending on sustainType
    if sustainType == 'mixture_GMM' or sustainType == "mixture_KDE": # Add labels column back
    on before returning (KDE uses labels col)
        data_preprocessed = np.concatenate((data_adjusted, array_of_clinical_labels[:,
        None]), axis=1)
        return data_preprocessed

    # ELIF NEEDS TESTED
    elif sustainType == "zscore": # Take z scores of data and return without labels column
        # Re-splitting updated data for step (2)
        array_of_clinical_labels = pd.DataFrame(array_of_clinical_labels)
        data_with_labels = pd.concat([data_adjusted, array_of_clinical_labels], axis=1)
        data_with_labels.columns = ["motsscore", "tfcscore", "mmsettotal", "irascore",
        "exfsscore", "labels"]
        # data = data_with_labels[data_with_labels['labels'].values == [1]].iloc[:, :5]
        data = data_with_labels[data_with_labels['labels'].values == [1]].iloc[:,
        :num_predictors]
        # data_controls = data_with_labels[data_with_labels['labels'].values == [0]].iloc[:,
        :5] # Subset controls for use in z score modelling
        data_controls = data_with_labels[data_with_labels['labels'].values == [0]].iloc[:,
        :num_predictors] # Subset controls for use in z score modelling

        # (2) Calculate the mean and standard deviation of each biomarker in your controls
        dataset
        mean_controls = np.mean(data_controls, axis=0)
        std_controls = np.std(data_controls, axis=0)

        # (3) Z-score your data by taking (data-mean_controls)/std_controls.
        data = (data - mean_controls) / std_controls
        data_controls = (data_controls - mean_controls) / std_controls

        # (4) Identify any biomarkers that decrease with disease progression, these will have
        mean_data < mean_controls.
        # Multiply the data for these biomarkers by -1.
        # *** NEEDS AUTOMATED
        IS_decreasing_1 = np.mean(data, axis=0) < np.mean(data_controls, axis=0)
        print(IS_decreasing_1)
        data = pd.DataFrame((data - mean_controls) / std_controls)
        data_controls = pd.DataFrame((data_controls - mean_controls) / std_controls)
        data.iloc[:, 0] = data.iloc[:, 0].mul(-1) # motsscore
        data.iloc[:, 2] = data.iloc[:, 2].mul(-1) # mmsettotal
        data.iloc[:, 3] = data.iloc[:, 3].mul(-1) # irascore
        data.iloc[:, 4] = data.iloc[:, 4].mul(-1) # exfsscore
        IS_decreasing_2 = np.mean(data, axis=0) < np.mean(data_controls, axis=0)
        print(IS_decreasing_2)
        if return_stats==False:
            return np.array(data)
        elif return_stats==True:
            # Check that the mean of the controls population is 0 (means covariate regression
            worked)
            print("\n", 'Mean of disease dataset is', '\n', np.mean(data,axis=0))
            # Check that the standard deviation of the whole dataset is greater than 1
            print("\n", 'Standard deviation of disease dataset is', '\n', np.std(data,axis=0))
            print("\n", 'Mean of controls is', '\n', np.mean(data_controls,axis=0))
            # Check that the standard deviation of the controls population is 1
            print("\n", 'Standard deviation of controls is', '\n',
            np.std(data_controls,axis=0))
            # Check that the mean of the whole dataset is positive
            # print(IS_decreasing_1)
            # print(IS_decreasing_2)
            return np.array(data)

```

6.1.5 Appendix 5: Feature analysis python code

```
# FEATURE SELECTION

from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score
import scipy.stats
from scipy.stats import ttest_ind
from scipy.stats import wilcoxon
import numpy as np
import pandas as pd

#####

# CONTENTS
# (1) Wilcoxon non-parametric feature selection
# (2) Random forest non-parametric feature selection
# (3) T-test parametric feature selection

#####

# ----- (1) WILCOXON FEATURE SELECTION -----

def feature_selection_wilcoxon(data_all_features, sign_lev = 0.05):
    data_all_features = data_all_features.iloc[:, 1:] # Skip subjid
    wilc_results = {} # Create dictionary to store p-values of each biomarker
    count = 0
    for column in data_all_features:
        controls = data_all_features[data_all_features["hdcat"] == 0]
        disease = data_all_features[data_all_features["hdcat"] == 1].iloc[:controls.shape[0],:] # making disease same length as controls (wilcoxon requires it)
        wilc_score = scipy.stats.wilcoxon(x=disease[column], y=controls[column])
        if wilc_score[1] < sign_lev:
            count = count + 1
            wilc_results[column] = [round(wilc_score[0],2), round(wilc_score[1],4)] # adding key and value to dict
    wilc_results = sorted(wilc_results.items(), key=operator.itemgetter(1)) # Sorting dict by t-stat (changes dict to list)
    wilc_results = collections.OrderedDict(wilc_results) # Converting list back to dict
    wilc_csv = open("wilc.csv", 'a')
    wilc_csv.write("feature[wilc-stat, p-val]" + '\n' + '\n')
    for i in wilc_results.keys():
        wilc_csv.write(i + str(wilc_results[i]) + '\n')
    wilc_csv.close()
    return wilc_results

# ----- (2) RAND. FOREST FEATURE SELECTION -----

def feature_selection_rf(data):
    # data = data.iloc[:, 7:] # Ignoring first 7 features (subjectID etc)
    data = data[data["hdcat"] != 2] # Removing pre-manifest so have only manifest disease and controls
    data.dropna(axis=1, how='all', inplace=True)
    data.fillna(data.mean(), inplace=True)

    # Split train, test set
    first75 = round(len(data)*0.75)
    training_set = data[0:first75]
    test_set = data[first75:len(data)]

    x_train = training_set.loc[:, training_set.columns != 'hdcat'] # Selecting all cols except hdcat
    y_train = training_set["hdcat"].round()

    x_test = test_set.loc[:, test_set.columns != 'hdcat']
    y_test = test_set["hdcat"].round()
```

```

# RANDOM FOREST CLASSIFIER
rfmodel = RandomForestClassifier(n_estimators=24)
rfmodel.fit(x_train,y_train)
rfpredictions = rfmodel.predict(x_test)
resultRfmodel = accuracy_score(y_test, rfpredictions)
print('Random Forest accuracy: ', resultRfmodel)

# RANDOM FOREST IMPORTANCES
feature_list = list(x_train.columns)
# Get numerical feature importances
importances = list(rfmodel.feature_importances_)
# List of tuples with variable and importance
feature_importances = [(feature, round(importance, 2)) for feature, importance in
                        zip(feature_list, importances)]
# Sort the feature importances by most important first
feature_importances = sorted(feature_importances, key = lambda x: x[1], reverse = True)
# Print out the feature and importances
[print('Variable: {:20} Importance: {}'.format(*pair)) for pair in feature_importances];
# importances = [('{:20} Importance: {}'.format(*pair)) for pair in feature_importances];
# print(importances)

# ----- (3) T-TEST FEATURE SELECTION -----
# 1. Find mean and SD of each biomarker
# 2. Do T - tests between disease and controls
# 3. Select features with biggest effect size (difference between groups)
import collections
import operator
def feature_selection_ttest(data_all_features, sign_leve=0.05): # Data and controls
    data_all_features = data_all_features.iloc[:, 1:] # Skip subjid
    ttest_results = {} # Create dictionary to store p-values of each biomarker
    count = 0
    for column in data_all_features:
        controls = data_all_features[data_all_features["hdcat"] == 0]
        disease = data_all_features[data_all_features["hdcat"] == 1]
        t_test_score = scipy.stats.ttest_ind(disease[column], controls[column],
            nan_policy='omit')
        if t_test_score[1] < sign_leve:
            count = count + 1
            ttest_results[column] = [round(t_test_score[0],2), round(t_test_score[1],4)] #
            adding key and value to dict
    ttest_results = sorted(ttest_results.items(), key=operator.itemgetter(1)) # Sorting dict
    by t-stat (changes dict to list)
    ttest_results = collections.OrderedDict(ttest_results) # Converting list back to dict
    ttest_csv = open("ttest_results.csv", 'a')
    ttest_csv.write("feature[t-stat, p-val]" + '\n' + '\n')
    for i in ttest_results.keys():
        ttest_csv.write(i + str(ttest_results[i]) + '\n')
    ttest_csv.close()
    return ttest_results

```

6.1.6 Appendix 6: Run file python code

Code adapted from: <https://github.com/ucl-pond/pySuStaIn/blob/master/simrun.py>

Calls functions from: <https://github.com/ucl-pond/pySuStaIn>

```
# pySuStaIn: SuStaIn algorithm in Python (https://www.nature.com/articles/s41468-018-05892-0)
# Author: Peter Wijeratne (p.wijeratne@ucl.ac.uk)
# Contributors: Leon Aksman (l.aksman@ucl.ac.uk), Arman Eshaghi (a.eshaghi@ucl.ac.uk), Alex
Young (alexandra.young@kcl.ac.uk)
```

```
#####
```

```
# CONTENTS
# (0) Define settings: Sustain type & features etc
# (1) Fix erroneous values
# (2) Select Baseline / Follow up only
# (3) Get rid of medical & demographic features (around 80 columns)+ drop columns w >50% NaNs
# (4) Normality test
# (5) Feature selection
# (6) Features subset
# (7) Deal with missing values
# (8) Remove outliers
# (9) Covariate adjustment(+ Z-scoring if sustainType=ZScoreSustain)
# (10) RUN model using preprocessed data
# (11) T-test differences between subtypes
```

```
#####
```

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
import os
from kde_ebm.mixture_model import fit_all_kde_models, fit_all_gmm_models
from kde_ebm import plotting
from matplotlib import cbook as cbook
import warnings
warnings.filterwarnings("ignore", category=cbook.mplDeprecation)
from ZscoreSustain import ZscoreSustain
from MixtureSustain import MixtureSustain
import sklearn.model_selection

from pkg0.preprocessing_pipeline import * # Created by author, Lucy Rothwell
from pkg0.feature_selection import * # Created by author, Lucy Rothwell

# PIPELINE: LOAD AND PRE-PROCESS DATA
enroll_df = pd.read_csv("/Users/lucyrothwell/Google_Drive/MSc_Comp/9_Dissertation HD/Data -
Enroll HD/enroll.csv", delimiter=',')
profile_df = pd.read_csv("/Users/lucyrothwell/Google_Drive/MSc_Comp/9_Dissertation HD/Data -
Enroll HD/profile.csv", delimiter=',')
data = re_code_hdcat(enroll_df) # Recoding the HD category column

# # ----- Histogram to check spread -----
# testing_var = "chorlue"
# df = enroll_df[enroll_df[testing_var] > -50]
# # df = df[df["hdcat"]==0]
# df = df[df[testing_var] < 150]
# tms = df[testing_var].values
# hist = []
# labels = []
# for i in np.unique(df['hdcat'].values):
#     if np.isnan(tms[df['hdcat']==i]).all():
#         continue
#     hist.append(tms[df['hdcat']==i])
#     labels.append(str(i))
# plt.hist(hist,label=labels,stacked=True)
# plt.yscale("log")
# plt.xlabel(testing_var)
# plt.ylabel('Frequency')
# plt.legend()
# plt.show()
# # -----
```



```

## (0) Define settings: Sustain type & features etc
sustainType = "mixture_KDE" # for use in functions later
num_covars = 2
features_list = ["subjid", "motscore", "chorrue", "sacimuth", "sdmt2", "verfct7", "trlb3",
"depscore", "irascore", "pbas3sv", "age", "iscad", "hdcat"] # 9 fits - TEST - USING FS VARS
num_predictors = len(features_list)-num_covars-2 # 2 to also take away hdcat and subjid

# (1) Fix erroneous values
data = fix_erroneous(data)

# (2) Select Baseline / Follow up only
data = data[data['visit'].values == ["Baseline"]] # Selecting BASELINE/FU visits only
data.drop_duplicates(subset="subjid", inplace=True, keep="first") # Removing duplicates

# (3) Get rid of medical & demographic features (around 80 cols) + drop cols with >50% NaNs
print("data(pre delete med_dem).shape:", data.shape)
data = delete_med_dem(data)
print("data(post delete med_dem).shape:", data.shape)

# > Make all values numeric
for column in data.iloc[:, 1:]: # Except subject id - col 0
    data[column] = pd.to_numeric(data[column])

print("Columns with > 40% NaNs (which have been dropped):")
for column in data.iloc[:, :-1]: # -1 to exclude isced
    if (data[column].isna().sum() / data.shape[0]) > 0.4:
        print(column)
        data.drop(column, inplace=True, axis=1)

# # (4) Normality test
# data_noNan = missing_vals(data, print_miss_counts=True, method="mean")
# for column in data_noNan.iloc[:, 1:]: # skipping subjid
#     title = column
#     column = data_noNan[column]
#     stat, p = scipy.stats.normaltest(column)
#     if stat < 300:
#         print(title, round(stat, 4), round(p,8))

# # (5) Feature selection
# wilcoxon_results = feature_selection_wilcoxon(data.iloc[:, 1:])
# rf_importances = feature_selection_rf(data.iloc[:, 1:])
# ttest_results = feature_selection_ttest(data)

# (6) Features subset
data = data[features_list]
# data[data["hdcat"]==1].hist()

# (7) Deal with missing values
data = missing_vals(data, print_miss_counts=True, method="drop")

# # -----(Counting analysis data for write-up)-----
# Merge data to get data counts
data_m = pd.DataFrame(data)
data_merge = pd.merge(data_m, profile_df, on="subjid", how='left')
# print(data_merge["hdcat"].value_counts(normalize=True)*100)
# print(data_merge["sex"].value_counts(normalize=True)*100)
# print(data_merge["region"].value_counts(normalize=True)*100)
# print(data_merge["race"].value_counts(normalize=True)*100)

# Getting CAG array to concatenate to output later
data_merge_no_preman = data_merge[data_merge["hdcat"] != 2]
caghigh = pd.to_numeric(data_merge_no_preman["caghigh"])
caglow = pd.to_numeric(data_merge_no_preman["caglow"])
type(caghigh.iloc[0])
row = 0
for i in caghigh:

```

```

        if i == 1:
            caghigh.iloc[row] = caglow.iloc[row]
            row = row + 1
cag = caghigh
# -----

# (8) Remove outliers
# > Remove outliers 5*SDs from mean
disease = data[data["hdcats"]==1]
controls = data[data["hdcats"]==0]
preman = data[data["hdcats"]==2]
if sustainType == 'mixture_GMM' or sustainType == "mixture_KDE": # Include hdcats (a few extra
steps for this)
    feat = features_list[1:-3] # # 1 for subjID
    feat.append(features_list[-1]) # Adding hdcats
    print("\n", "DISEASE")
    disease = outlier_removal(disease, feat, sd_num=5) # 1 to skip subject ID
    print("\n", "CONTROLS")
    controls = outlier_removal(controls, feat, sd_num=5) # 1 to skip subject ID
    print("\n", "PREMAN")
    preman = outlier_removal(preman, feat, sd_num=5) # 1 to skip subject ID
    data = pd.concat([disease, controls, preman])
    elif sustainType == 'zscore': # Exclude hdcats.
    data = outlier_removal(data, features_list[1:-3], sd_num=5) # 1 for subjID still in

# -----

# (9) Covariate adjustment (+ Z-scoring if sustainType=ZScoreSustain)
# Regresses out covariates of predictors + if ZScoreSustain; (2) Finds mean and SD of
features, (3) Takes Z scores, (4) Makes sure all Z scores are increasing
data = covariate_adjustment(data.iloc[:, 1:], sustainType, num_predictors)
# Check adjustment has worked - i.e., mean of each control biomarker == 0.
feat = 0
for column in data[data[:, -1] == 0].T: # Check mean of each control biomarker now == 0
(means adjustment has worked)
    mean = np.mean(column)
    print("control mean, biomarker", feat, "=", np.around(mean, 4))
    feat = feat + 1

# # **** Do this if not doing covariate adjustment- to get rid of covariates ****
# data = pd.DataFrame(data)
# hdcats = data.iloc[:, -1]
# predictors = data.iloc[:, 1:len(features_list)-num_covars-1] # 1 to exclude subjID, 2
covariates and hdcats
# data = pd.concat([predictors, hdcats], axis=1)
# data = np.array(data)
# -----

#----- HISTS CHECK -----
# data_hists = pd.DataFrame(data)
# data_hists = data_hists.iloc[:, :-1]
# data_hists.columns = ["motscore", "chorrue", "sacinith", "sdmt2", "verfct7", "trlb3",
"depscore", "irascore", "pbas3sv"]
#
# # data_hist[data_hist.iloc[:, -1] == 1].hist()
# # data_hist[data_hist.iloc[:, -1] == 0].hist()
# # print(motscore.value_counts())
# data_hists.hist()
# -----

# -----end of pre-processing-----

# (10) RUN model using preprocessed data
def main(data, sustainType):
    # cross-validation
    validate = True
    N_folds = 10 # recommended 10

    if sustainType == "mixture_KDE" or "mixture_GMM":
        N = data.shape[1]-1 # number of biomarkers - if kdemmm - hdcats is still in data so we

```

```

    need to -1 to get num of biomarkers
if sustainType == "zscore":
    N = data.shape[1] # number of biomarkers

print("N (1) in main()=", N)

M = data.shape[0] # number of observations ( e.g. subjects )

N_startpoints = 25 # Number of starting points to use when fitting the subtypes
    hierarchichally. I'd suggest using 25.
N_S_max = 3 # The maximum number of subtypes to fit. I'd suggest starting with a lower
    number - maybe three - and then increasing that if you're getting a significantly
    better fit with the maximum number of subtypes. You can judge this roughly from the
    MCMC plot. To properly evaluate the optimal number of subtypes you need to run
    cross-validation.

N_iterations_MCMC = int(1e5) # The number of iterations for the MCMC sampling of the
    uncertainty in the progression pattern. I'd recommend using 1x10^5 or 1x10^6.
# N_iterations_MCMC = 100 # Use this one to make programme quicker on test runs. CHANGE
    back toint(1e5) for final run.

# either 'mixture_GMM' or 'mixture_KDE' or 'zscore'
sustainType = sustainType

assert sustainType in ("mixture_GMM", "mixture_KDE", "zscore"), "sustainType should be
    either mixture_GMM, mixture_KDE or zscore"

dataset_name = 'enroll.csv'
output_folder = dataset_name + '_' + sustainType

Z_vals = np.array([1, 2, 3] * N) # Z-scores for each biomarker
Z_max = np.array([5] * N) # maximum z-score

if sustainType == 'mixture_GMM' or sustainType == "mixture_KDE":

    case_control_all = data[data[:, -1] != 2] # Removing pre-manifest
    labels_case_control = case_control_all[:, -1] # getting labels (-1 to remove hdcatt)
    data_case_control = case_control_all[:, :-1] # getting predictors (-1 to remove
        hdcatt)
    data = data_case_control
    print("data.shape", data.shape)
    print("labels_case_control.shape", labels_case_control.shape)

    N = data.shape[1] #hdcatt already removed so no need for-1
    print("N (2) in main()=", N)

    SuStaInLabels = []
    SuStaInStageLabels = []

    for i in features_list[1:num_predictors+1]:
        SuStaInLabels.append(i)

    print("SuStaInLabels", SuStaInLabels)

# FIT THE BIOMARKER DATA - FINDING A "CLUSTER" FOR DISEASE AND A "CLUSTER" FOR CONTROLS
# I.e., Estimate the parameters (mean and variance) of the Gaussian (GMM) or estimate the
    non-parametric (KDE)
if sustainType == "mixture_GMM":
    mixtures = fit_all_gmm_models(data, labels_case_control.astype(int))
elif sustainType == "mixture_KDE":
    # mixtures = fit_all_kde_models(data, labels_case_control.astype(int))
    mixtures = fit_all_kde_models(data, labels_case_control.astype(int))

outdir = "/Users/lucyrothwell/Google_Drive/MSc_Comp/9_Dissertation HD/SuStaIn-
    master/"

fig, ax = plotting.mixture_model_grid(data_case_control, labels_case_control,
    mixtures, SuStaInLabels)
fig.show()
fig.savefig(os.path.join(outDir, 'kde_fits.png')) #outdir should be output_folder

L_yes = np.zeros(data.shape)
L_no = np.zeros(data.shape)

# Calculating data likelihood for each biomarker, N:
for i in range(N):
    # BREAKING mixtures LIKELIHOODS INTO NORMAL AND ABNORMAL (based on likelihoods)

```

```

        if sustainType == "mixture_GMM":
            L_no[:, i], L_yes[:, i] = mixtures[i].pdf(None, data[:, i])
        elif sustainType == "mixture_KDE":
            L_no[:, i], L_yes[:, i] = mixtures[i].pdf(data[:, i].reshape(-1, 1)) #
        CALCULATING PDF
        # L_yes means likelihoods for abnormal
        sustain = MixtureSustain(L_yes, L_no, SuStaInLabels, N_startpoints, N_S_max,
                                N_iterations_MCMC, output_folder,
                                dataset_name, use_parallel_startpoints=True)

    elif sustainType == 'zscore':

        sustain = ZscoreSustain(data, Z_vals, Z_max, SuStaInLabels, N_startpoints, N_S_max,
                                N_iterations_MCMC,
                                output_folder, dataset_name, False)

    # RUNNING SUSTAIN ON OUTPUT OF KDEMM / GMM
    ml_subtype = sustain.run_sustain_algorithm()

    # CROSS VALIDATION
    if validate: # DO CROSS VALIDATION. A way of optimising for number of clusters.
        # Output here (cross_validated_sequences) will be messier. Check there is differences
        # between 1 cluster, 2 clusters etc.
        test_idx = []

        cv = sklearn.model_selection.StratifiedKFold(n_splits=N_folds, shuffle=True)
        # cv_it = cv.split(data, labels)
        cv_it = cv.split(data, labels_case_control.astype(int))

        for train, test in cv_it:
            test_idx.append(test)
        test_idx = np.array(test_idx)

        CVIC, loglike_matrix, y_arr_test = sustain.cross_validate_sustain_model(test_idx)

        # this part estimates cross-validated positional variance diagrams
        for i in range(N_S_max):
            sustain.combine_cross_validated_sequences(i + 1, N_folds)

    plt.show()
    return y_arr_test, ml_subtype

if __name__ == '__main__':
    np.random.seed(42)
    # main(data, sustainType) # (Original. Replaced with line below).
    y_arr_test = main(data, sustainType)

## (11) T-test differences between subtypes

print(ttest_ind(y_arr_test[0], y_arr_test[1]))
print(ttest_ind(y_arr_test[1], y_arr_test[2]))

t_test_12, p_val_12 = ttest_ind(y_arr_test[1], y_arr_test[2])
t_test_23, p_val_23 = ttest_ind(y_arr_test[1], y_arr_test[2])

p_val_12 = round(int(p_val_12), 4)
p_val_23 = round(int(p_val_23), 4)

## (12) T-tests between each subtype's CAG distribution

ml_subtype = pd.DataFrame(ml_subtype)
ml_subtype_wCAG = pd.concat([ml_subtype, cag], axis=1)
# ml_subtype_wCAG = ml_subtype_wCAG.dropna()

# print(ml_subtype_wCAG.iloc[:, 0].value_counts(normalize=True)*100)

sub1_cag = ml_subtype_wCAG.iloc[:, 1][ml_subtype_wCAG.iloc[:, 0]==0]
sub2_cag = ml_subtype_wCAG.iloc[:, 1][ml_subtype_wCAG.iloc[:, 0]==1]
sub3_cag = ml_subtype_wCAG.iloc[:, 1][ml_subtype_wCAG.iloc[:, 0]==2]

```

```
print(ttest_ind(sub1_cag, sub2_cag, nan_policy="omit"))  
print(ttest_ind(sub2_cag, sub3_cag, nan_policy="omit"))  
print(ttest_ind(sub1_cag, sub3_cag, nan_policy="omit"))
```